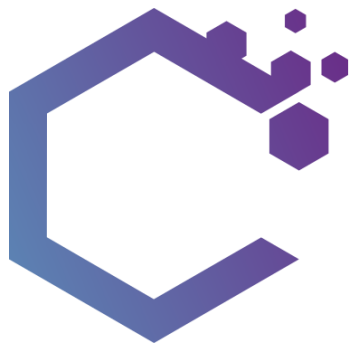




CEISAM

Chimie et Interdisciplinarité
Synthèse, Analyse,
Modélisation



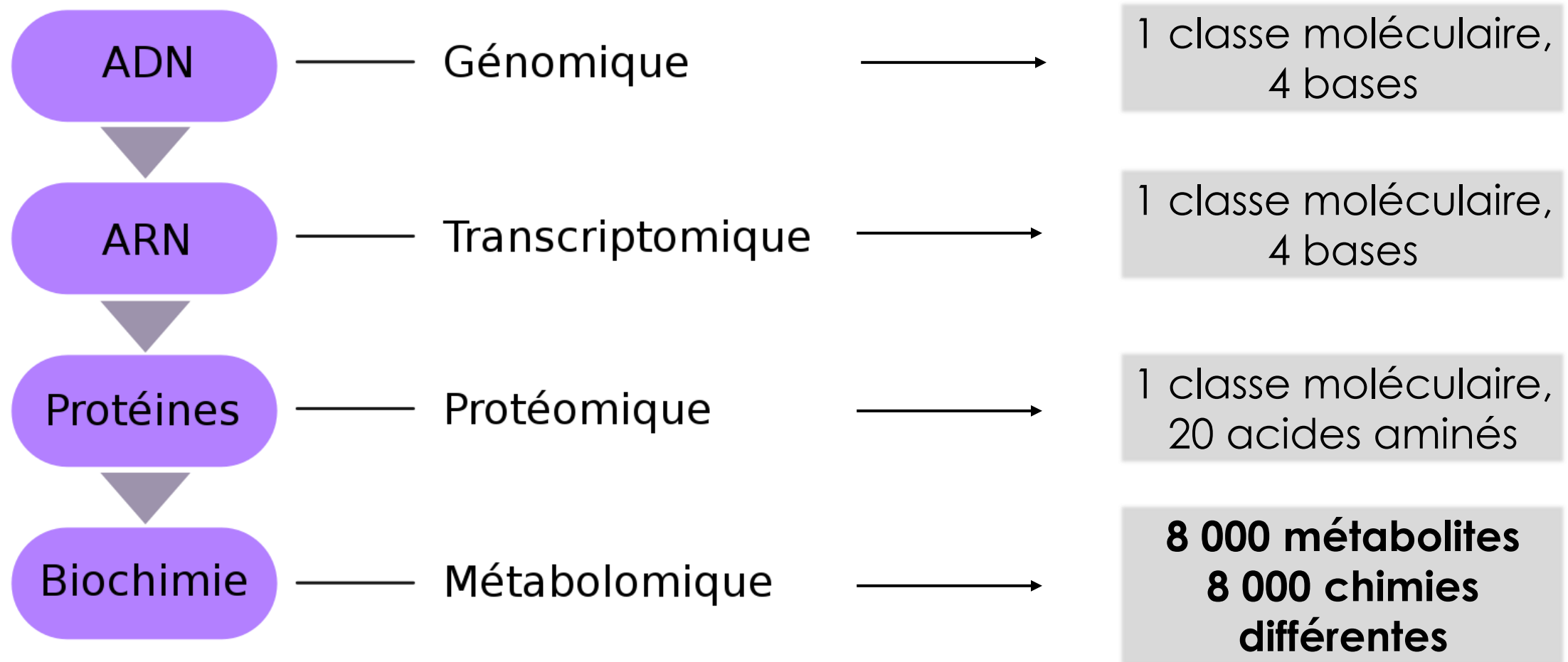
Nantes
Université



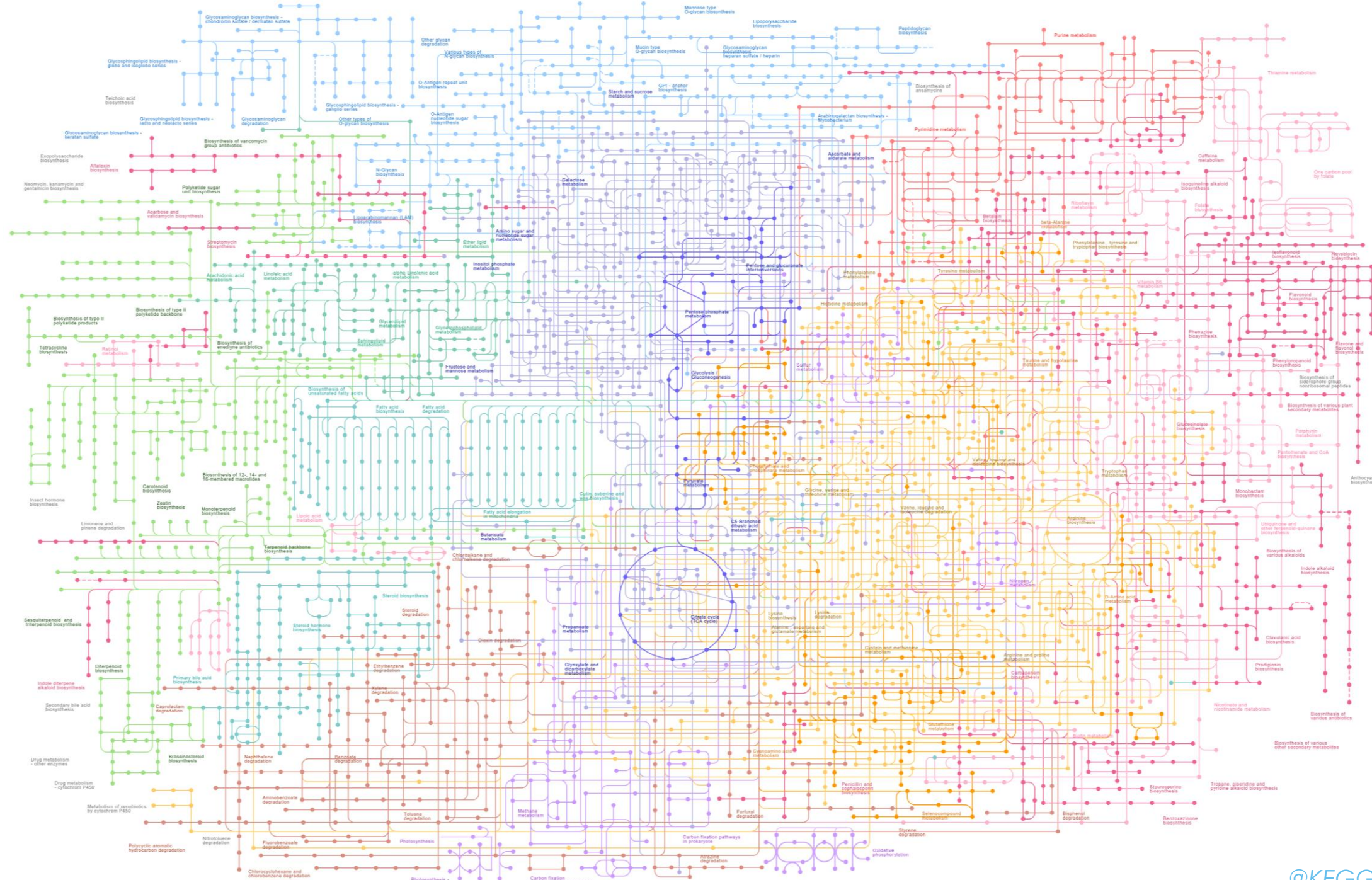
NMR and MS: complementary approaches

Thematic School #2 – RFMF 07/10/2026

Difficultés de la métabolomique



- ❖ Grand diversité chimique
- ❖ Ne peut être déduite du génome
- ❖ Grands écarts de concentrations (7 log)



Plateformes analytiques

NMR
spectroscopy
 ^1H mainly

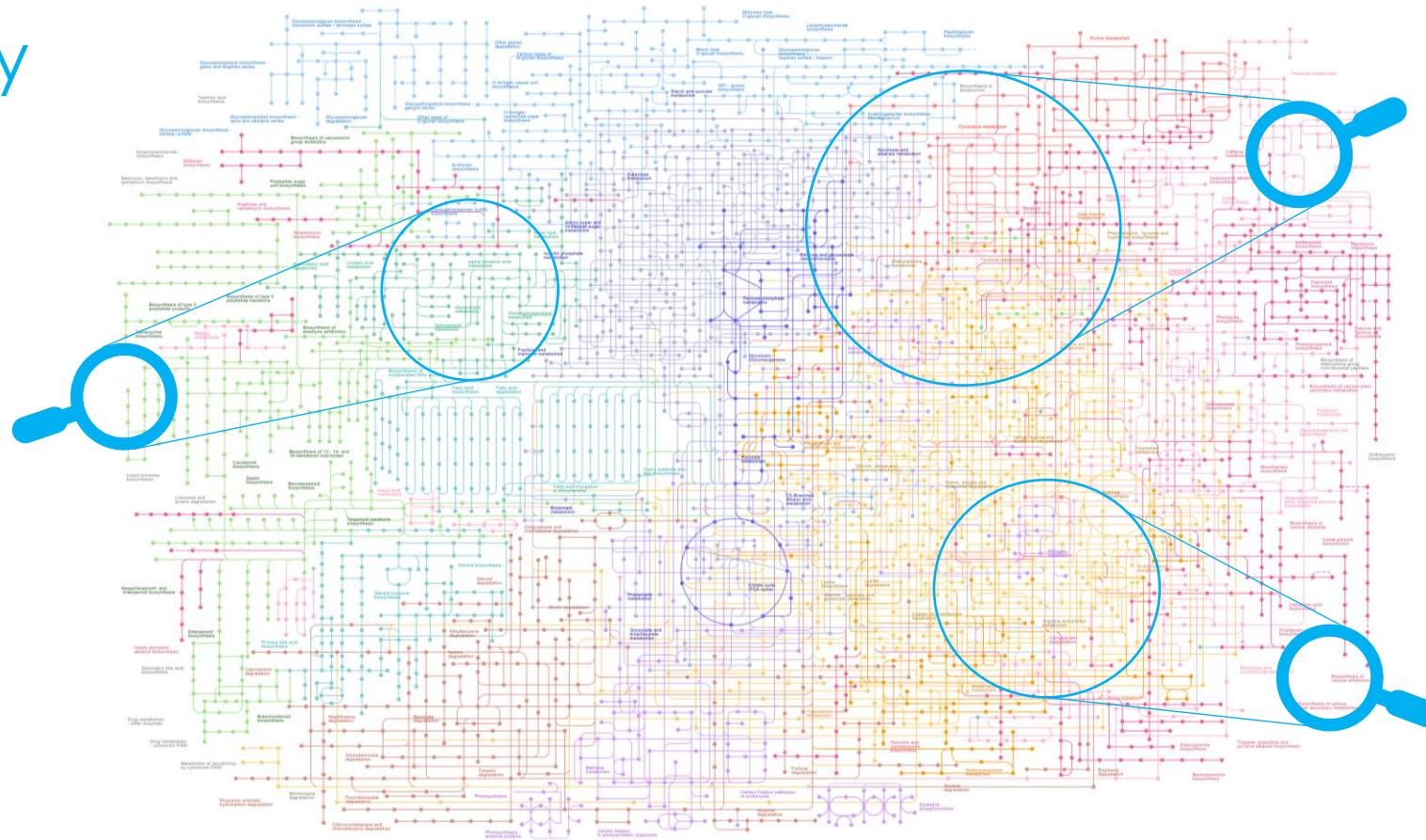


@CEISAM

LC-MS



@Waters



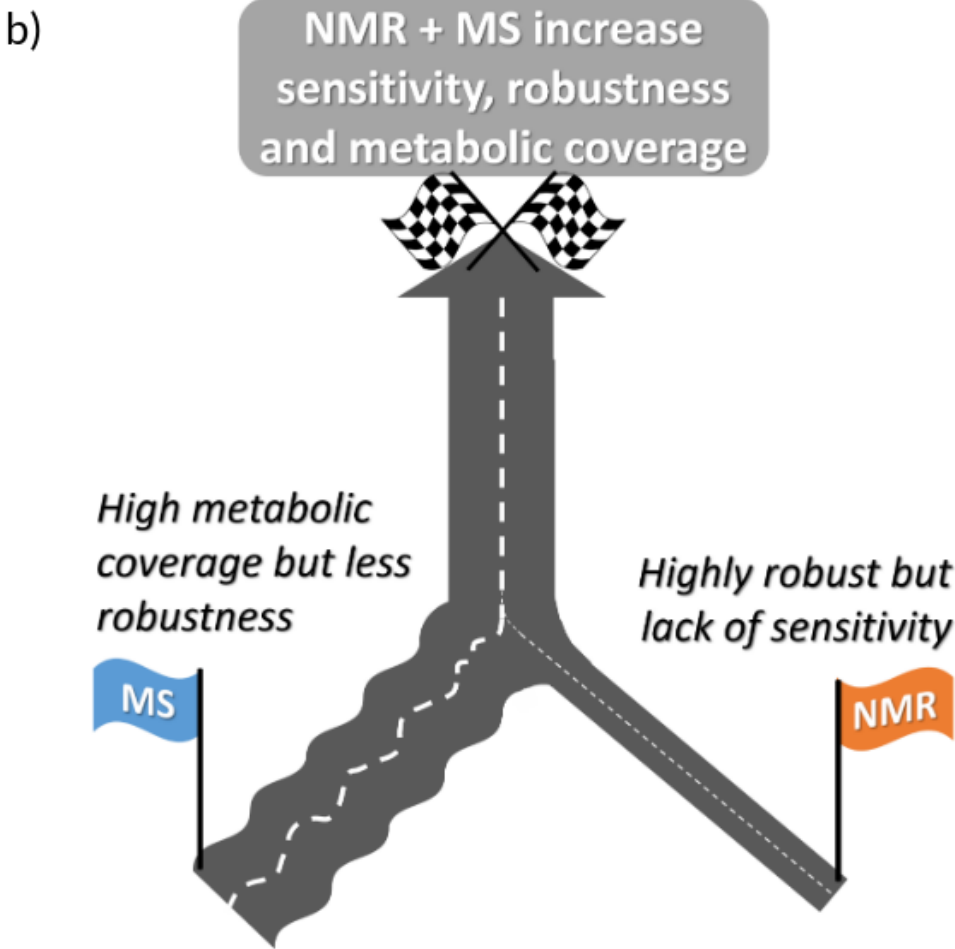
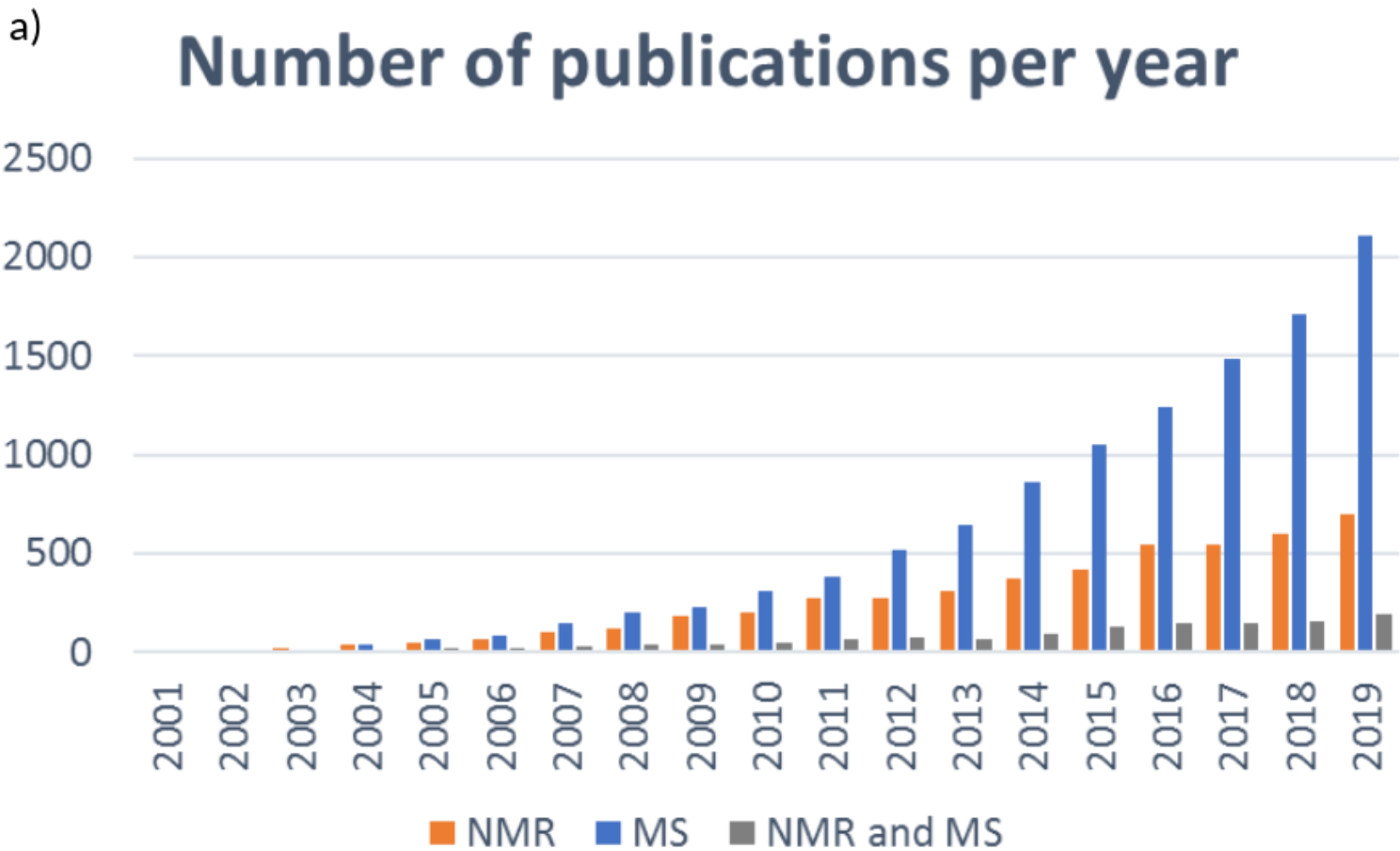
GC-MS



@ThermoFisher

MAS-NMR, Capillary Electrophoresis-MS, Ion Mobility-MS....

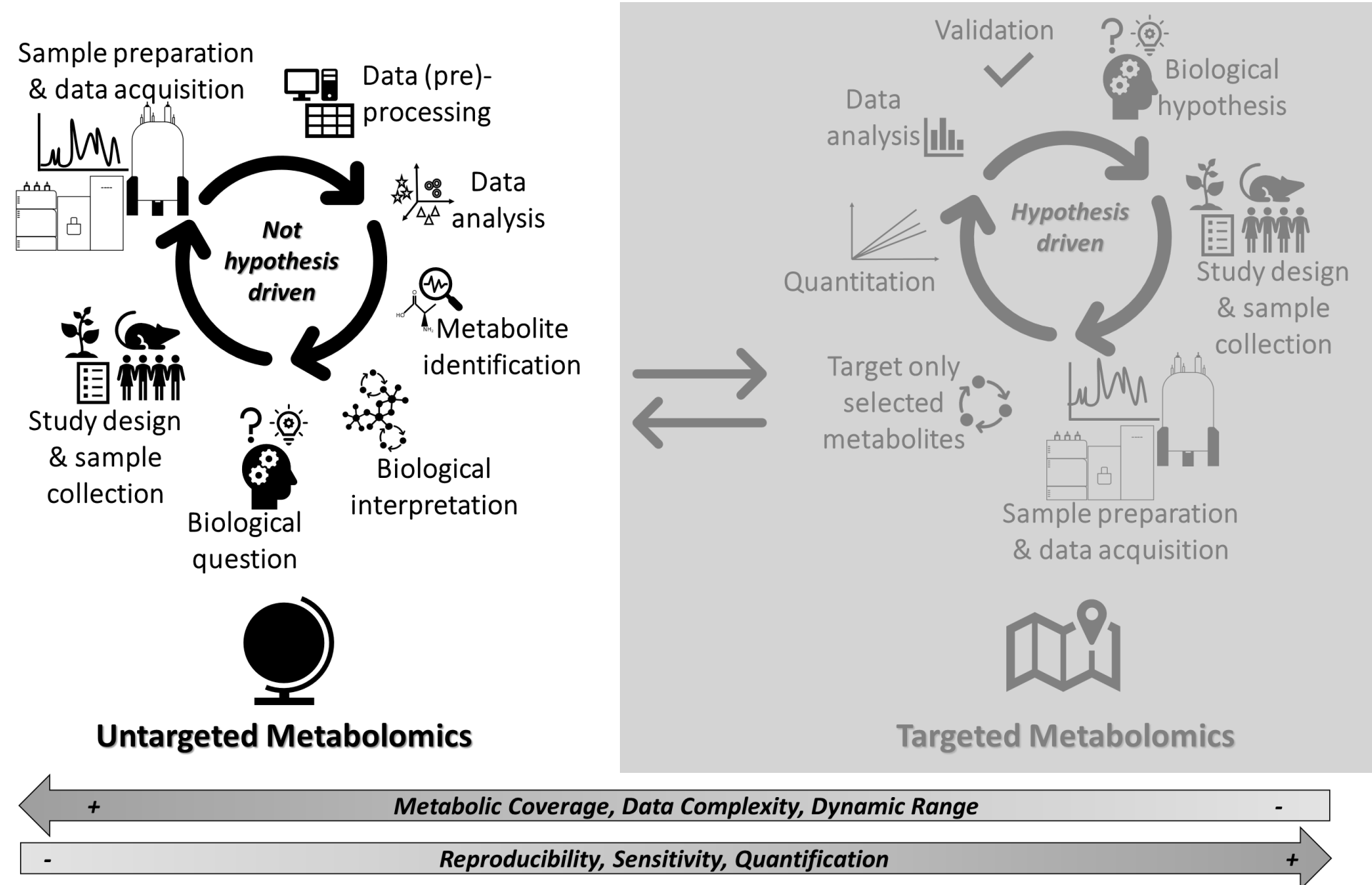
Le derby



C'est quoi une plateforme analytique idéale ?

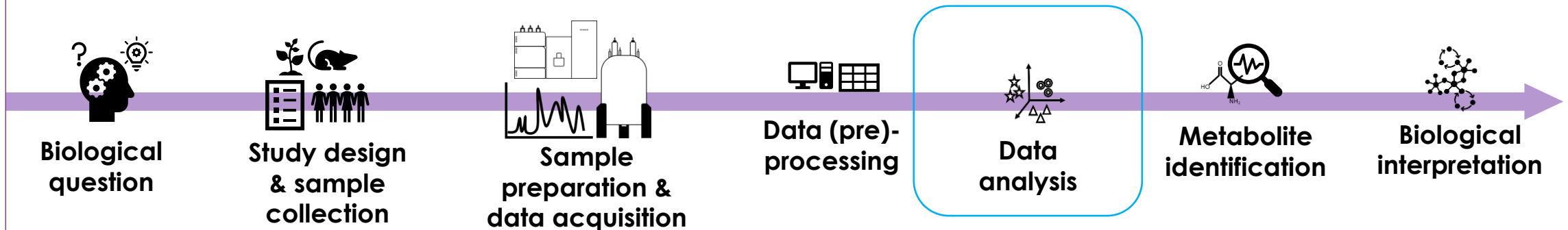
Qualities sought	NMR	GC-MS	LC-MS
Cost-limited instrument wise	More expensive and bigger	Cheapest and smaller	Cheaper and smaller
Simple sample preparation	Simple	The most complex (derivatization)	More complex
Non-destructive	Non destructive but large volume required	Destructive but small volume required	Destructive but small volume required
Cost-limited acquisition wise	Fine	Higher	Higher
Wide dynamic range	Low (10^3)	High (10^6)	High (10^6)
Acquisition rapidity	1D fast	Longer	Longer
Sensitivity	Much lower but can be improved (NS, cryoprobes, hyperpolarization)	High	Highest
Large metabolic coverage	Moderate (eg. CH, OA, AA...)	High (volatile, low to mid-polarity, eg. CH, OA, AA, FAs...)	Highest (mid to high-polarity, eg. CH, OA, AA, lipids...)
Matrix effect	NA	Limited	Important
Robustness and Reproducibility	Very high	Moderate	Poor
Easy automation (high-throughput)	Easiest	Easy	Easy
Easy metabolite identification	Relatively easy (well reported but limited by signal overlap)	Easy (well reported)	Difficult
Peak overlap	Important	Moderate	Moderate
Quantitative	1D with 1 SIL (linear response)	With SILs and CC (long)	With SILs and CC (long)
Structural information	Important (even more with 2D)	Limited	Limited

Dans le contexte d'une approche non-ciblée



Combined NMR spectroscopy and MS applied to metabolomics

- Data sets combination
 - Cross-comparison of NMR and MS data sets to increase metabolic coverage
 - Correlation of NMR and MS data sets
 - Multiblock fusion / integration



Aims and challenge of metabolite identification

What are you aiming for?

Annotate only the signals that appear to be discriminant after chemometrics statistical analyses

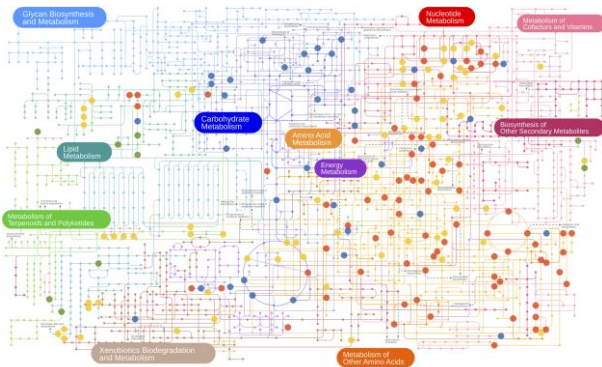
Annotate the whole spectrum to have the global picture of the physiology

Biological interpretation – place into context

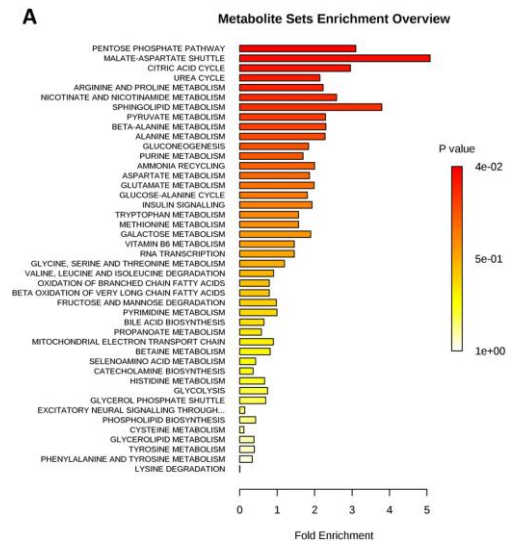
Pathway mapping

Functional Enrichment Analysis

Biomarker identification / Validation of a Diagnostic Biomarker



Wieder et al., *PLoS Comput. Biol.*, 2021



Fanelli et al., *Cell death Dis.*, 2018

Aims and challenge of metabolite identification

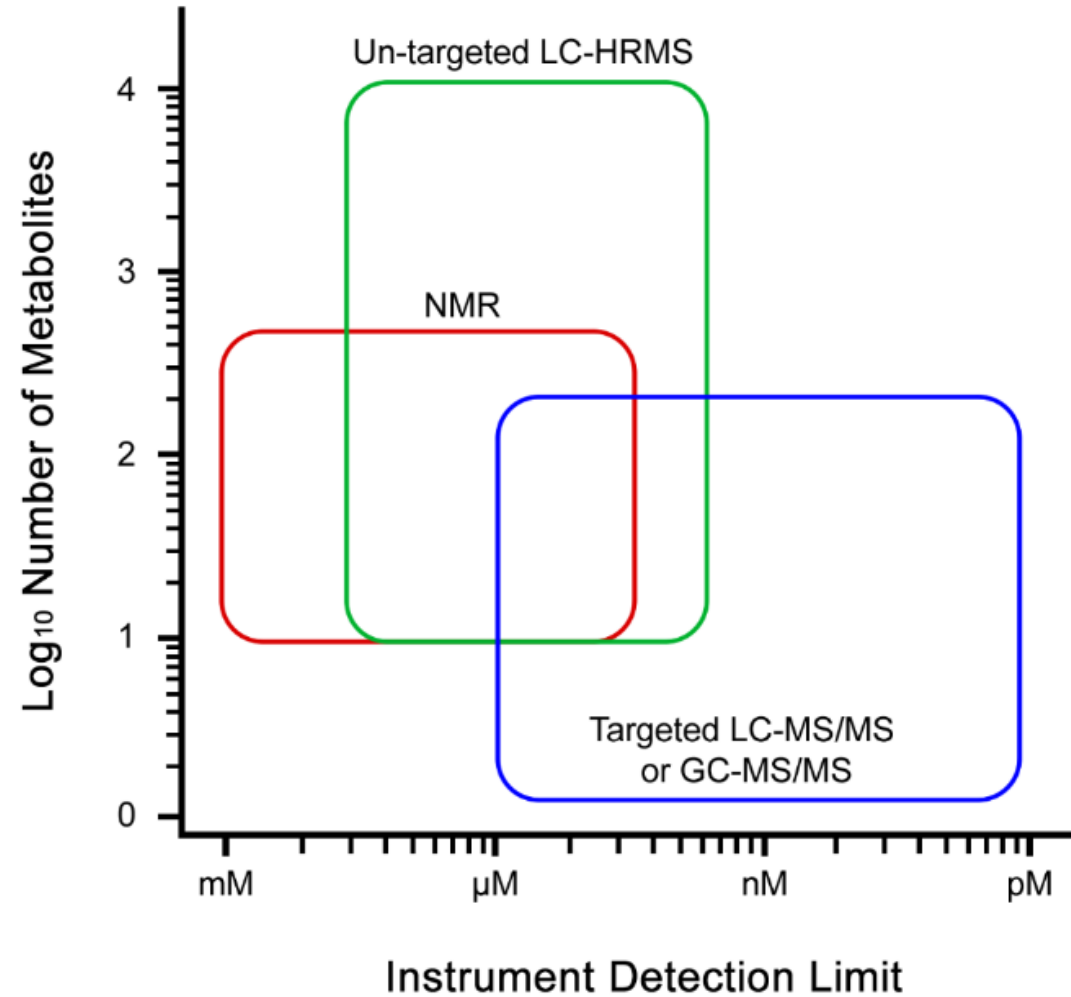
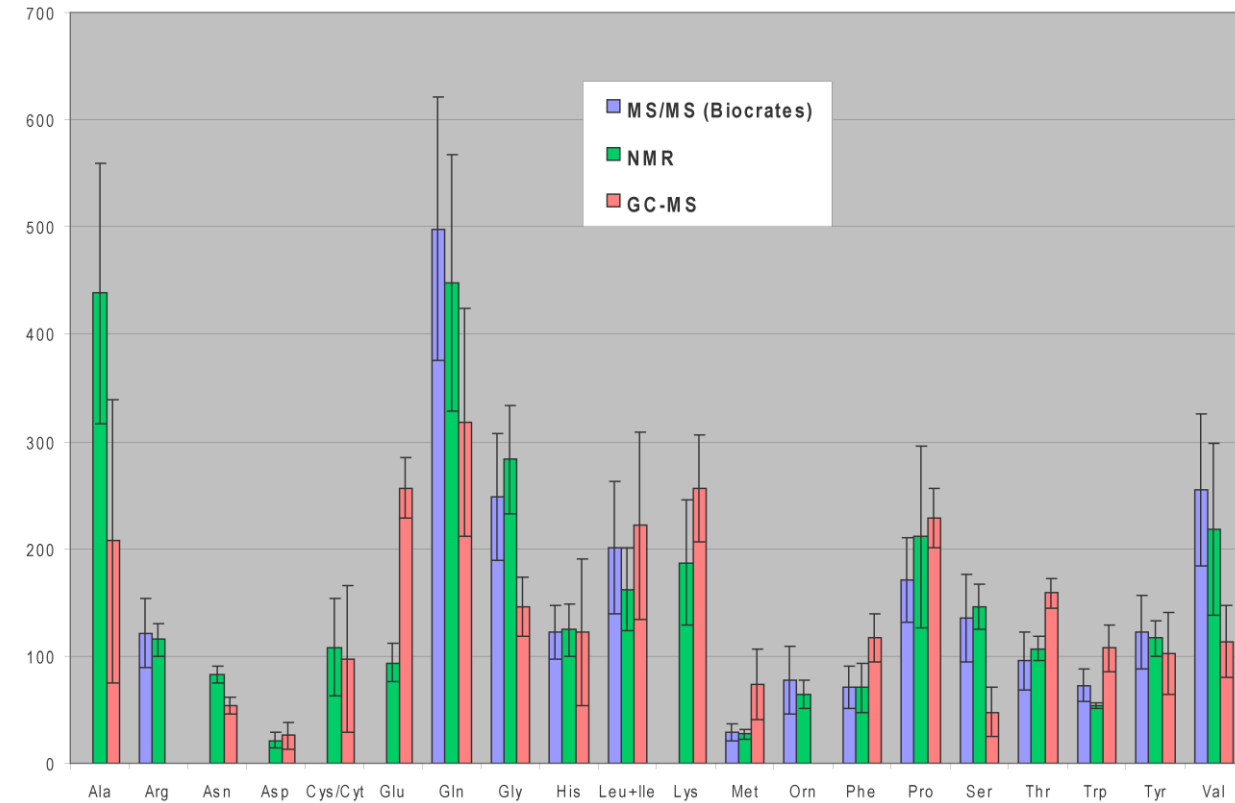
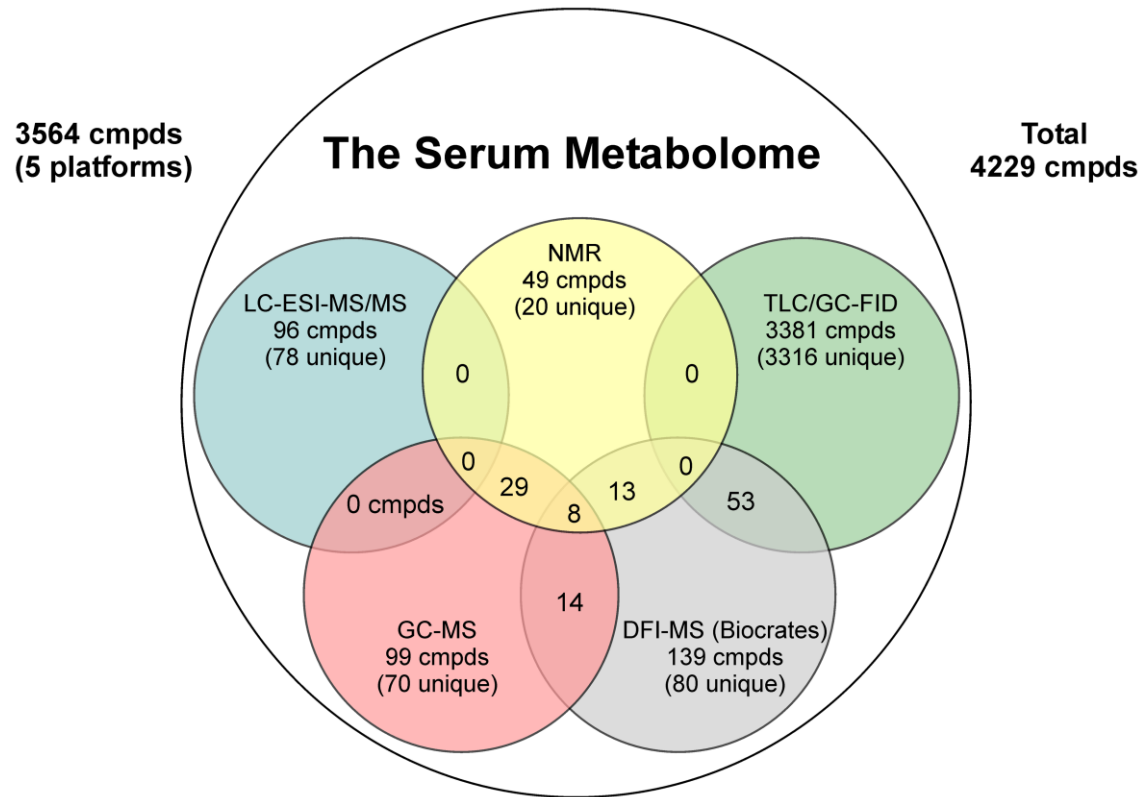


Figure 1. Schematic diagram showing the typical analysis platforms used for metabolomics experiments, illustrating the range of detection limits and number of detectable metabolites typically achieved.

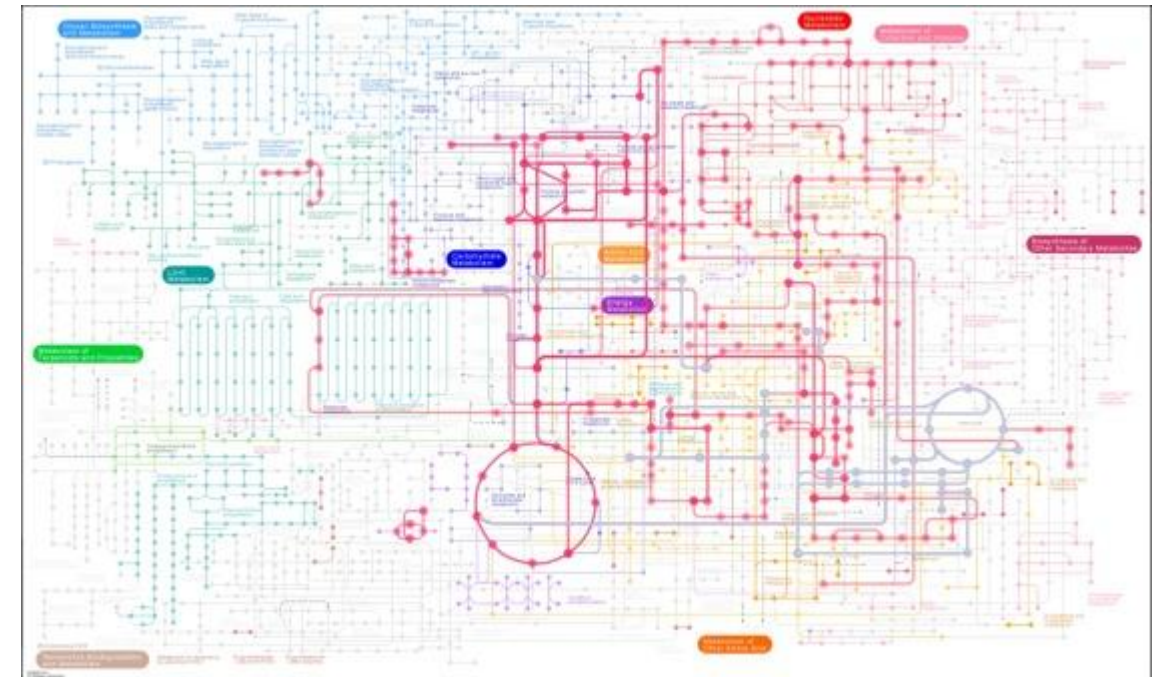
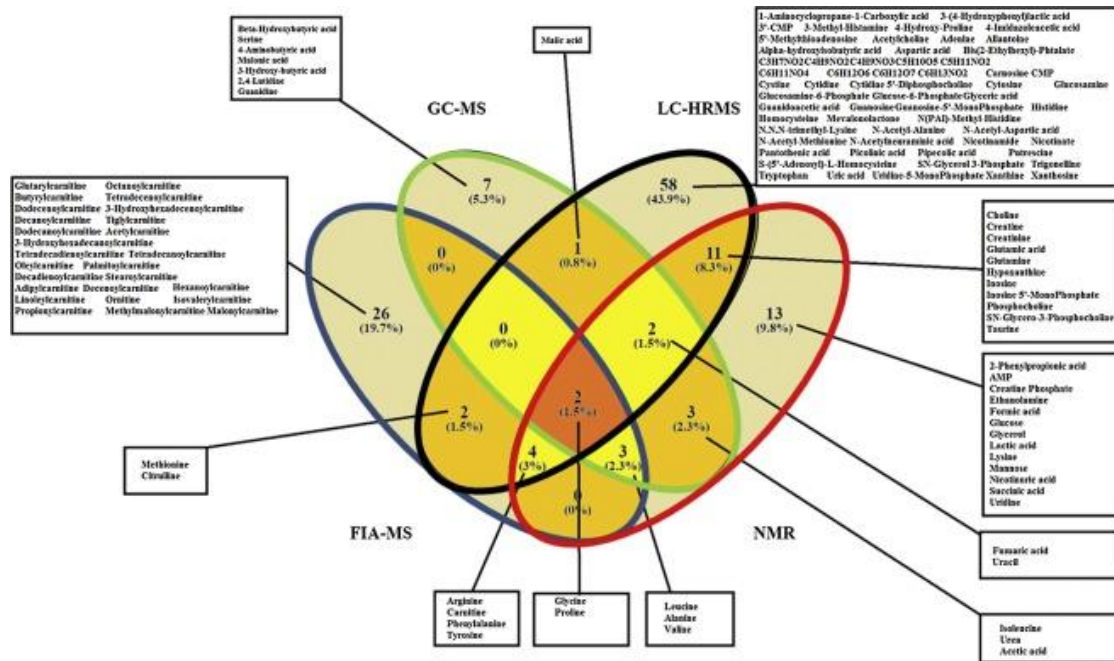
Data sets combination

Cross-comparison of NMR and MS data sets to **increase metabolic coverage**

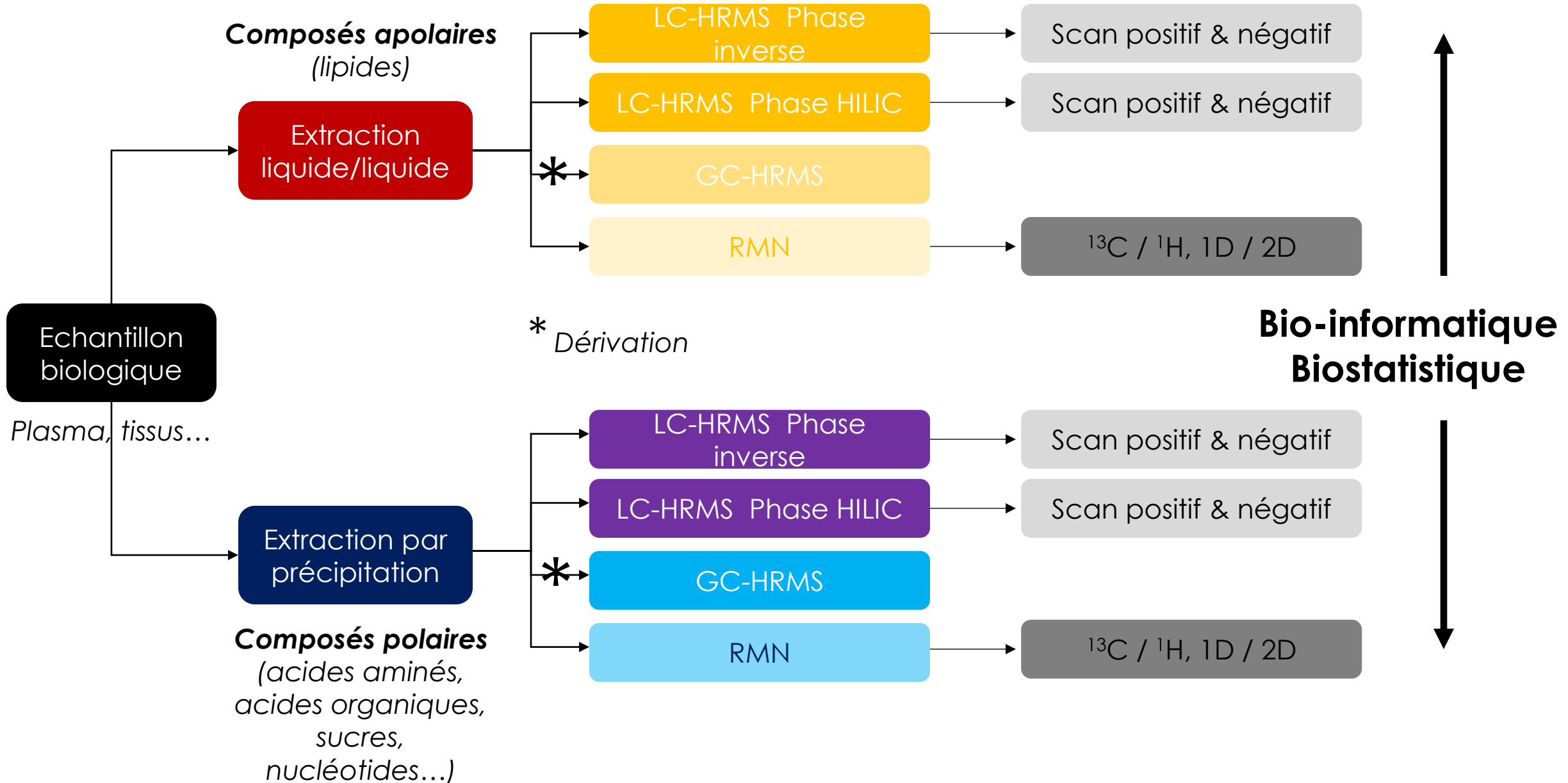


Data sets combination

Cross-comparison of NMR and MS data sets to **increase metabolic coverage**

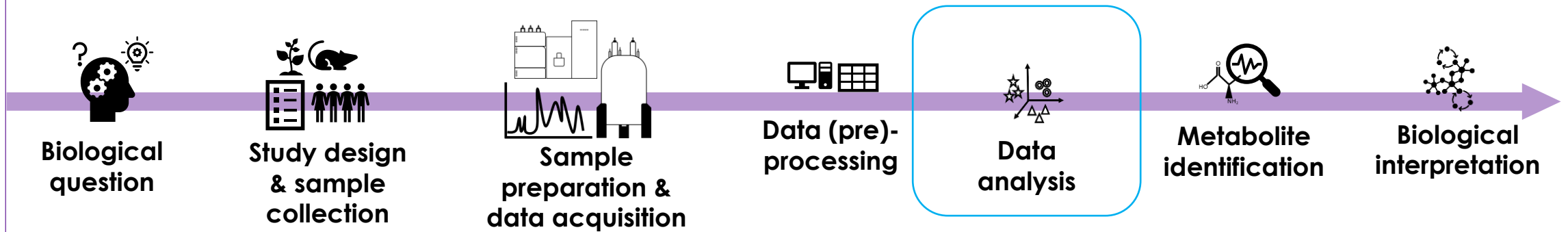


L'idéal ça serait...



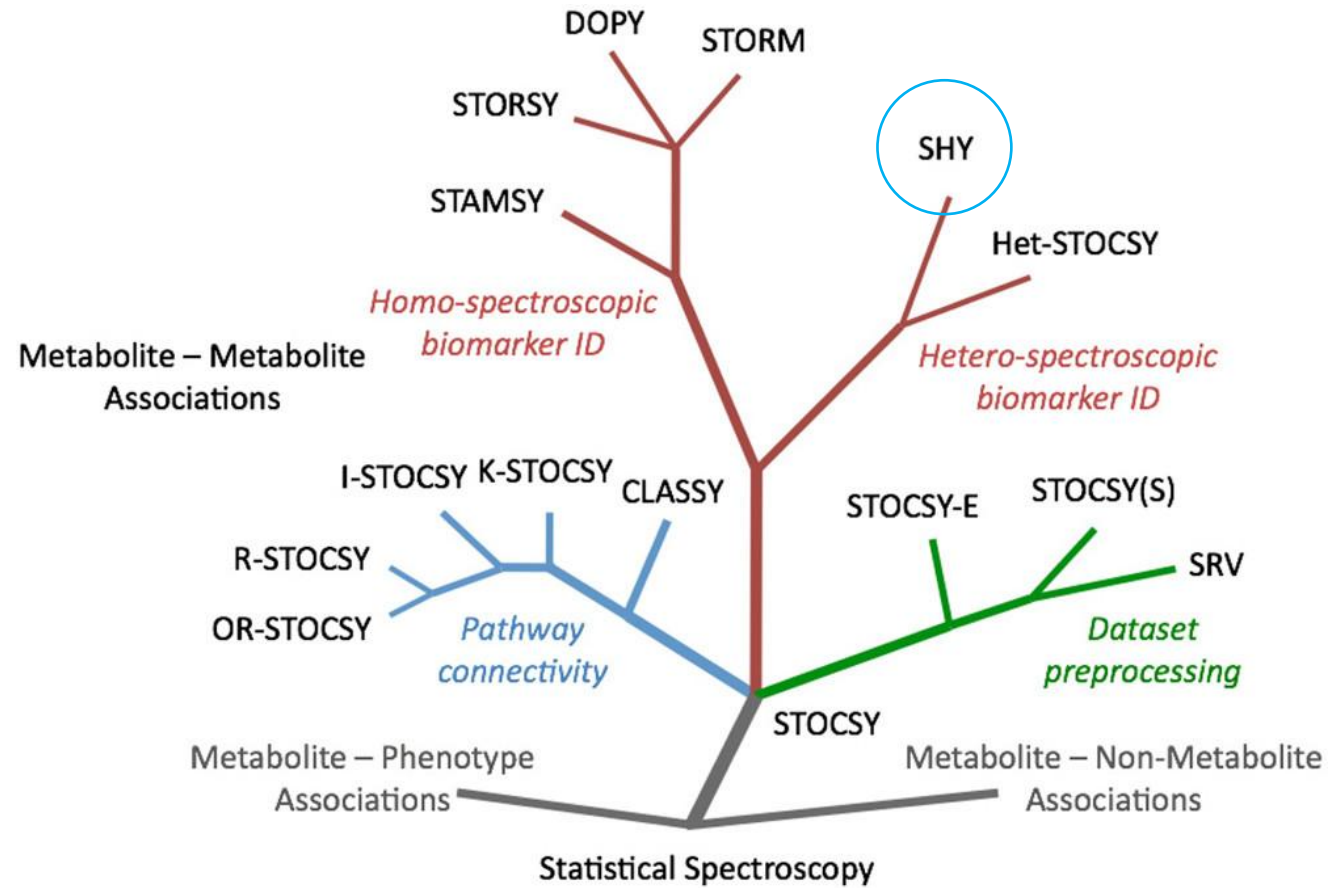
Combined NMR spectroscopy and MS applied to metabolomics

- Data sets combination
 - Cross-comparison of NMR and MS data sets to increase metabolic coverage
 - Correlation of NMR and MS data sets
 - Multiblock fusion / integration



Data sets combination

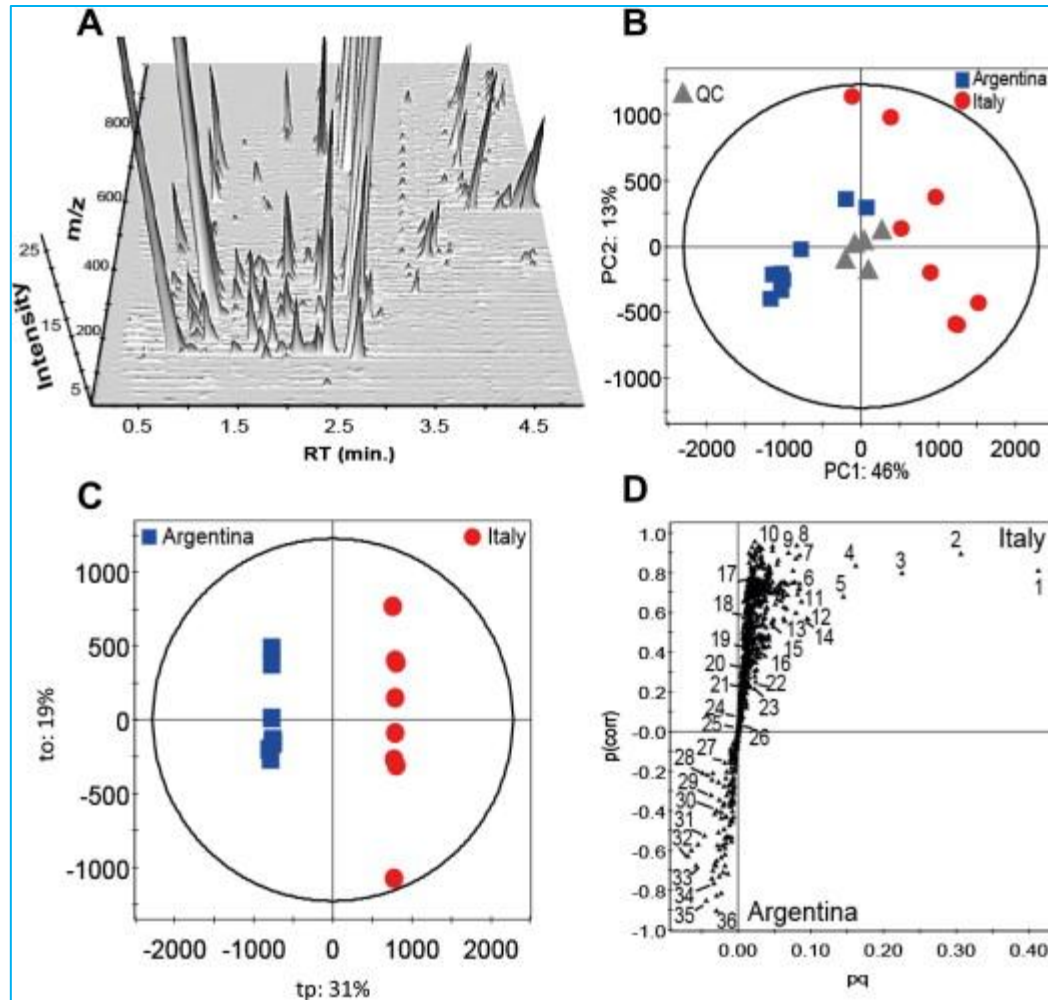
Correlation of NMR and MS data sets



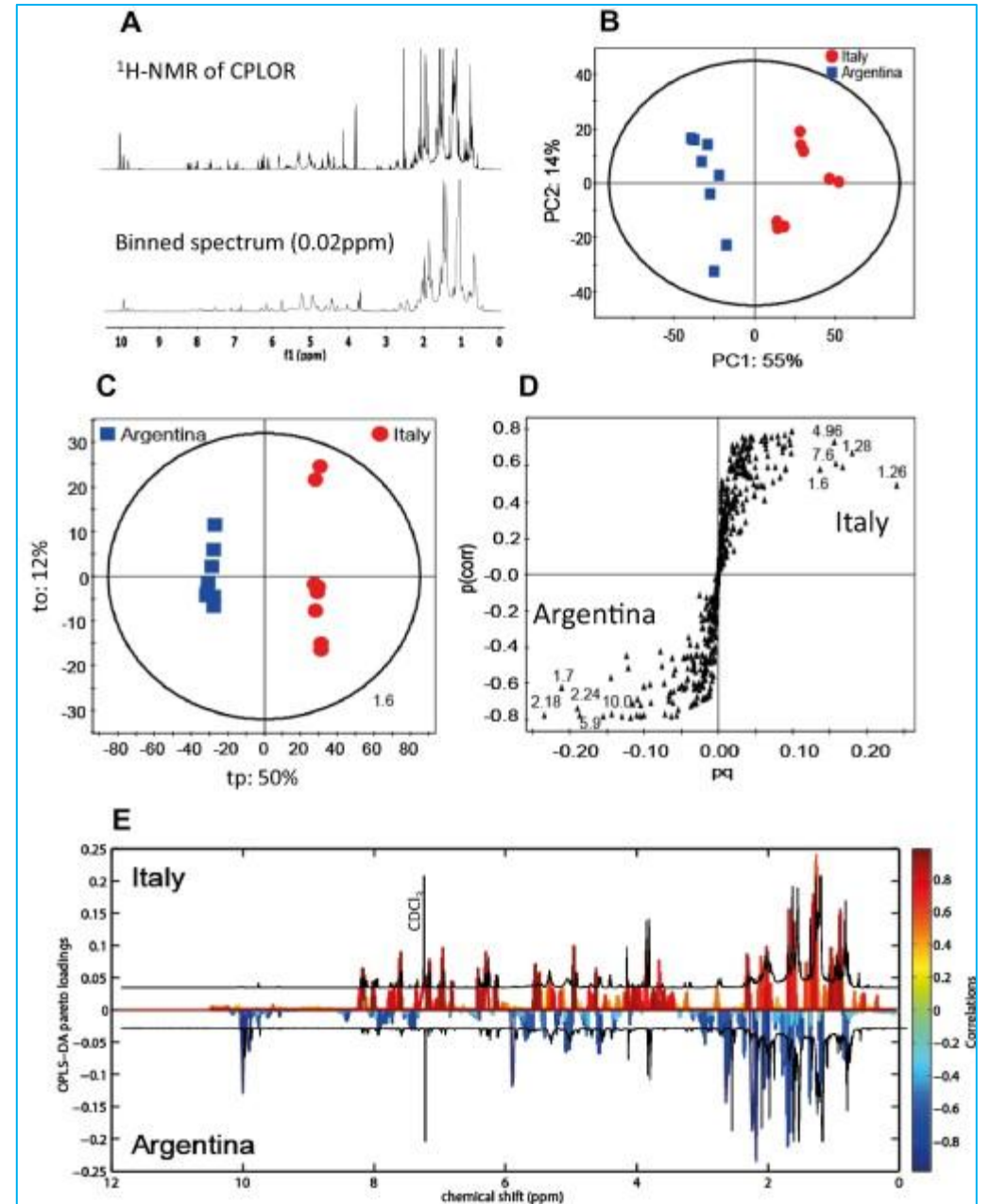
Data sets combination

Correlation of NMR and MS data sets

MS

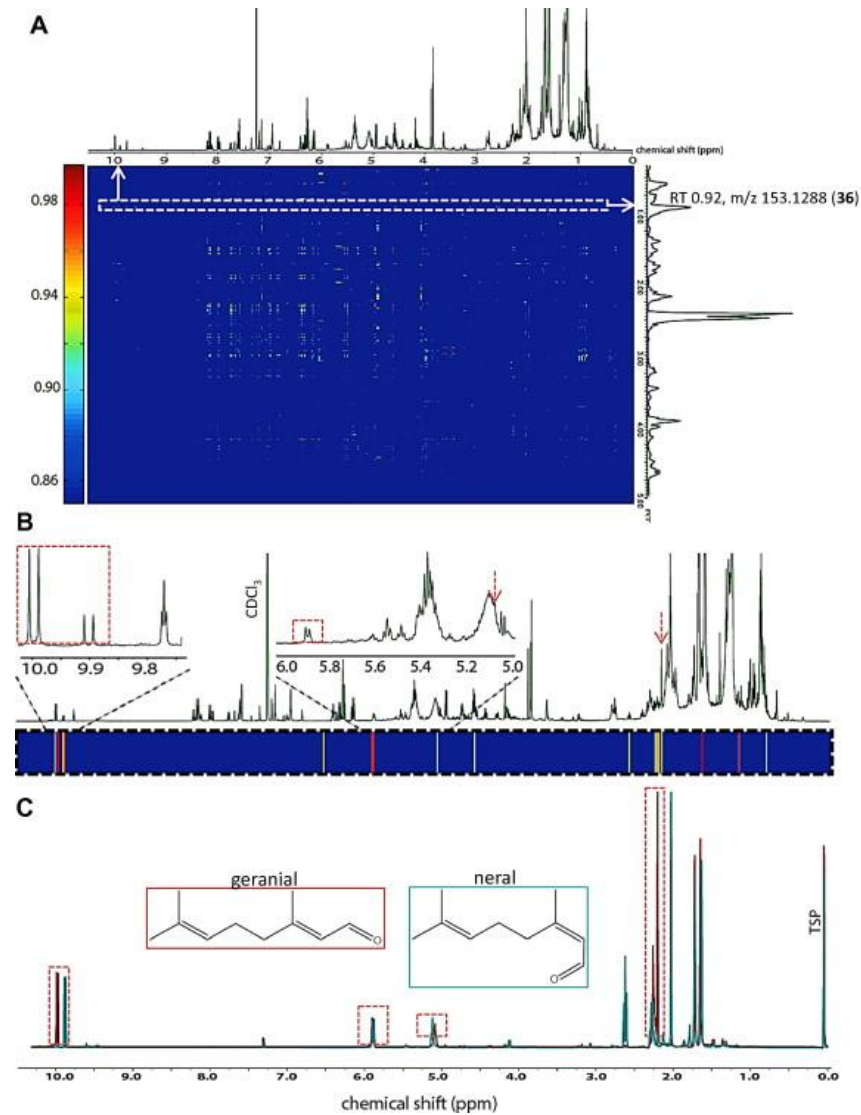


RMN



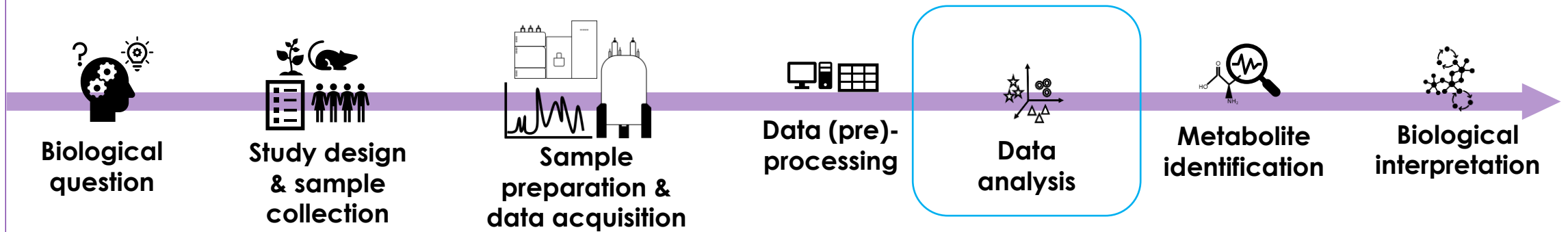
Data sets combination

Correlation of NMR and MS data sets



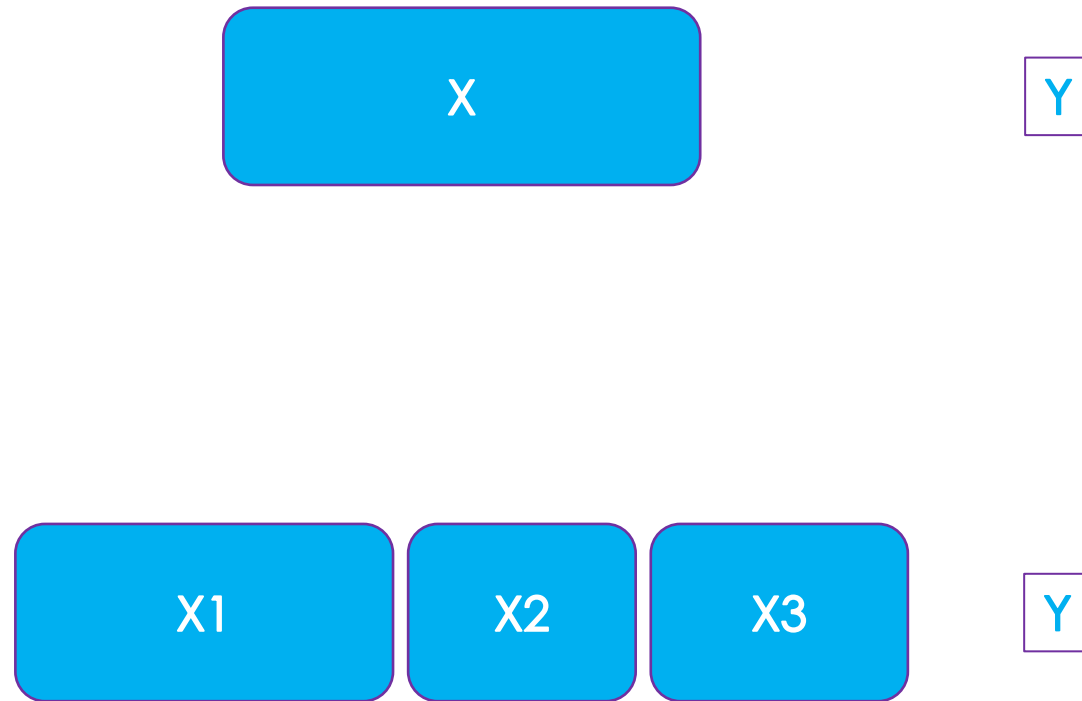
Combined NMR spectroscopy and MS applied to metabolomics

- Data sets combination
 - Cross-comparison of NMR and MS data sets to increase metabolic coverage
 - Correlation of NMR and MS data sets
 - Multiblock fusion / integration



Data sets combination

Multiblock fusion / integration



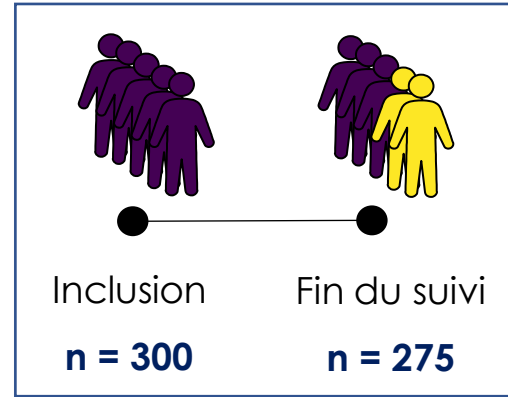
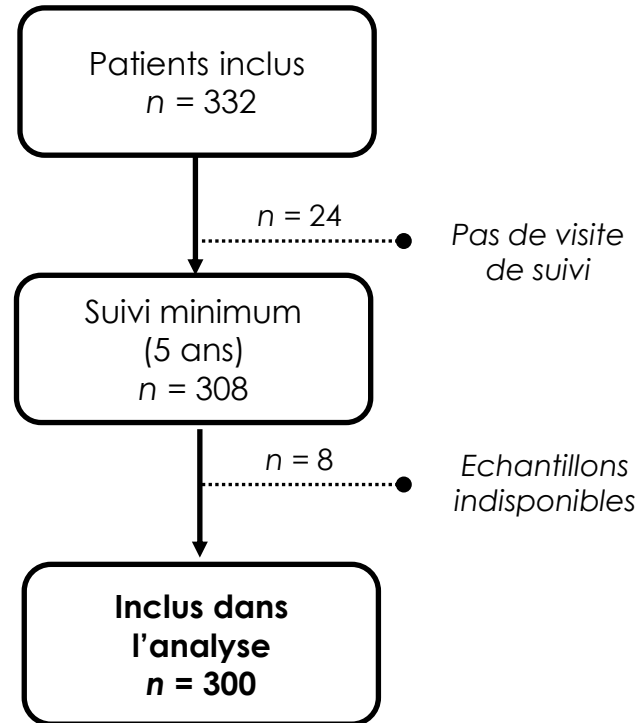
Classical: one high dimension matrix against a class matrix

Several high dimension matrices (eg. GC-MS, LC-MS and NMR) against a class matrix

IT-DIAB

Identifier de nouveaux biomarqueurs de risque de diabète chez les patients pré-diabétiques

Cohorte IT-DIAB



37 % de patients qui basculent (NOD+)

63 % de patients qui restent pré-diabétiques (NOD-)

Analyses multi-omiques

✓ Métabolomique

✓ Lipidomique

✓ Proteomique

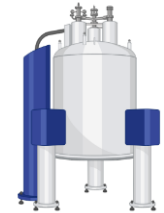
 M·shark



✓ Lipoproteomique

✓ Métabolomique

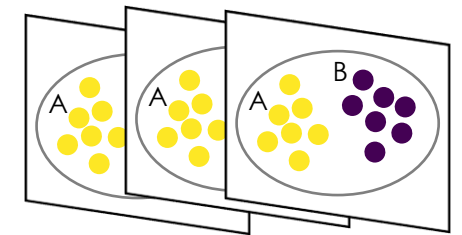
 CEISAM
Chimie et Interdisciplinarité
Synthèse, Analyse,
Modélisation



- ✓ Critère d'inclusion
 $110 < \text{GAJ} < 125 \text{ mg/dL}$
- ✓ Critère de jugement
 $\text{GAJ} \geq 125 \text{ mg/dL}$ ou glycémie $\geq 200 \text{ mg/dL}$ (2h post HGPO)

Analyses statistiques

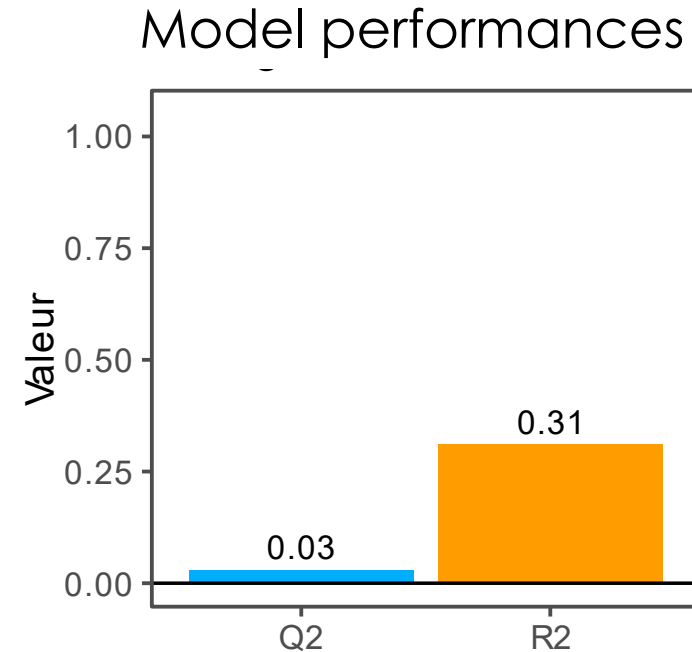
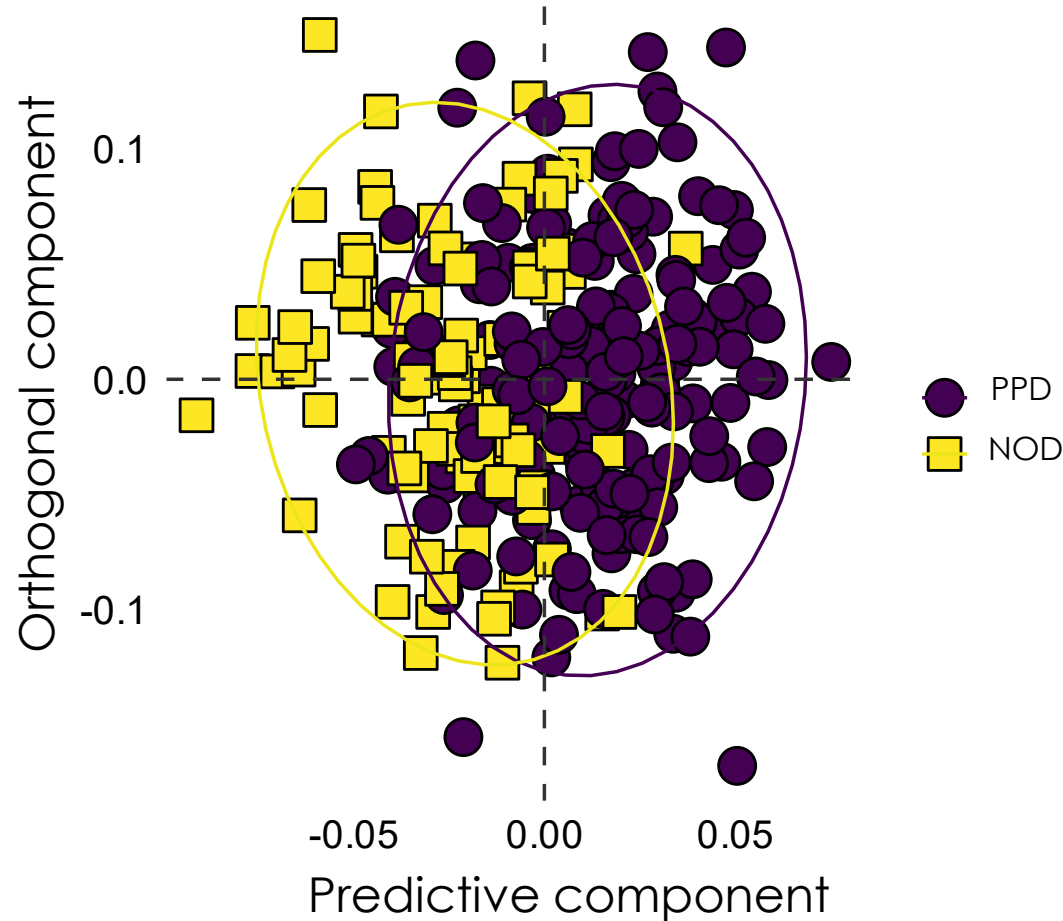
- ✓ Analyses multivariées
- ✓ Analyses univariées
- ✓ Corrélations de Spearman
- ✓ Kaplan-Meier
- ✓ Modèles de Cox



Intégration multi-block

Analyse multi-omique et pré-sélection des biomarqueurs

Samples collected at baseline (Vi) analysis

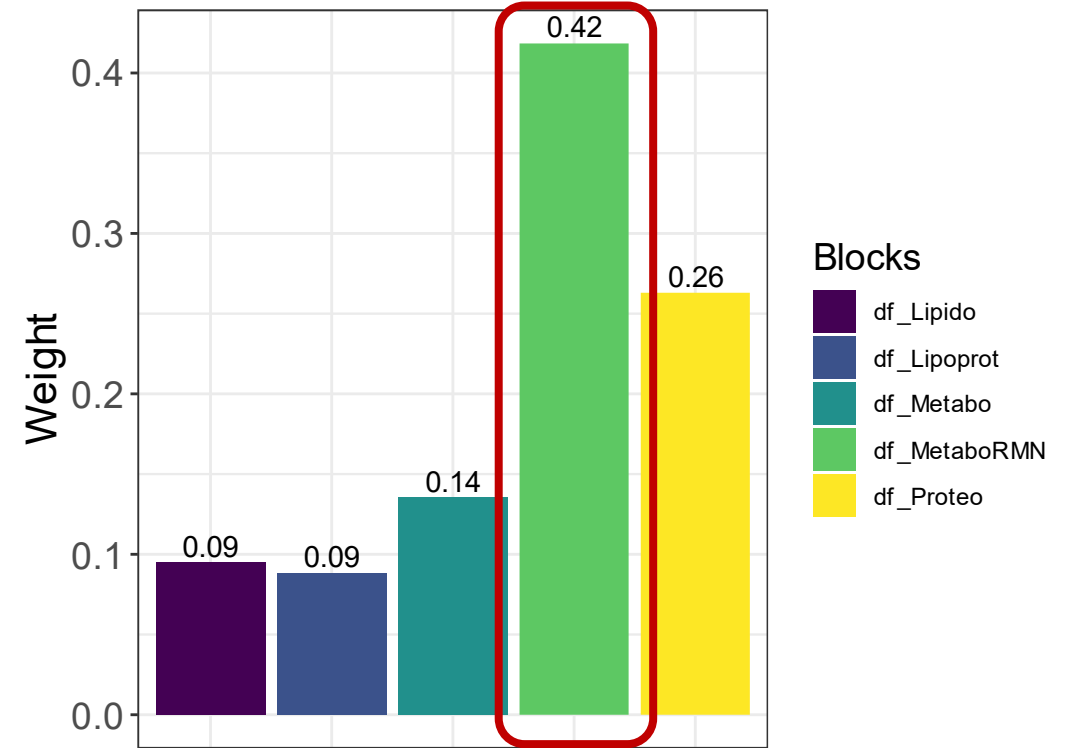
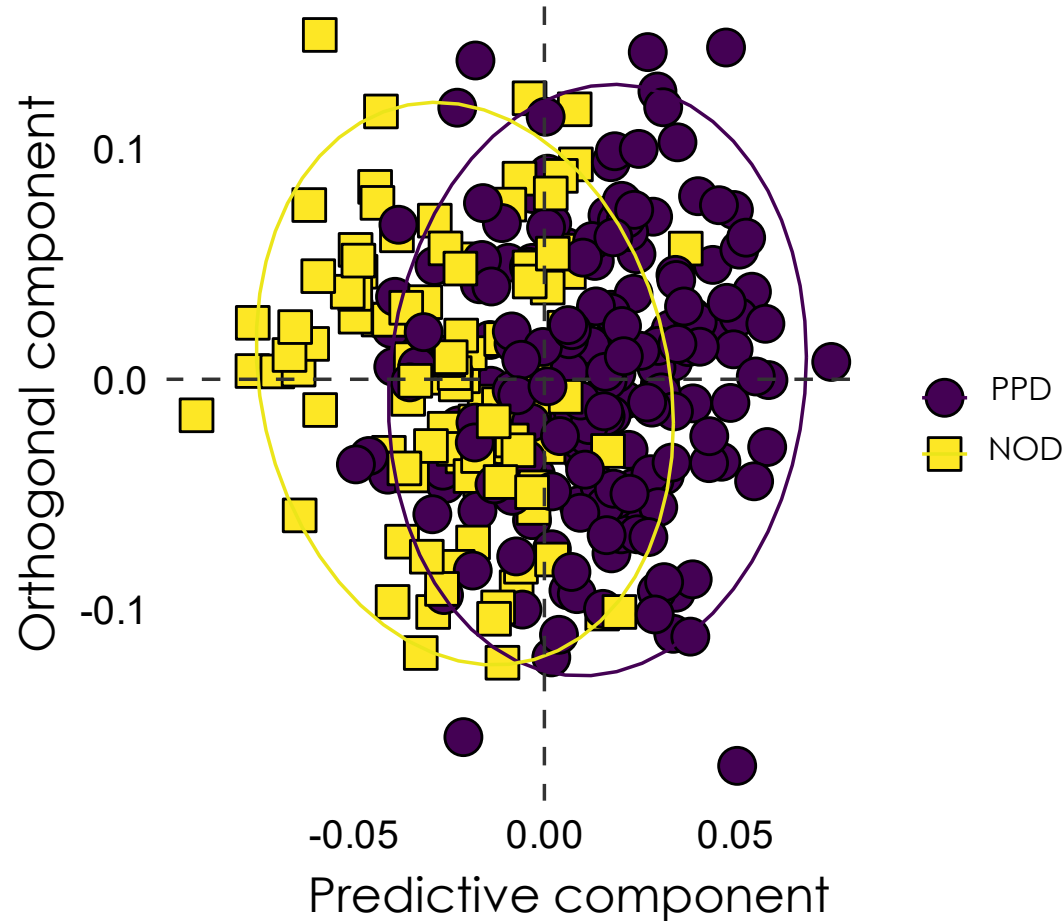


→ Poor performance due to the numerous variables not related to the development of T2D

779 variables
N-fold (n=10) cross validation₂₁
p value = 0.002 (Permutation test, n=500)

Analyse multi-omique et pré-sélection des biomarqueurs

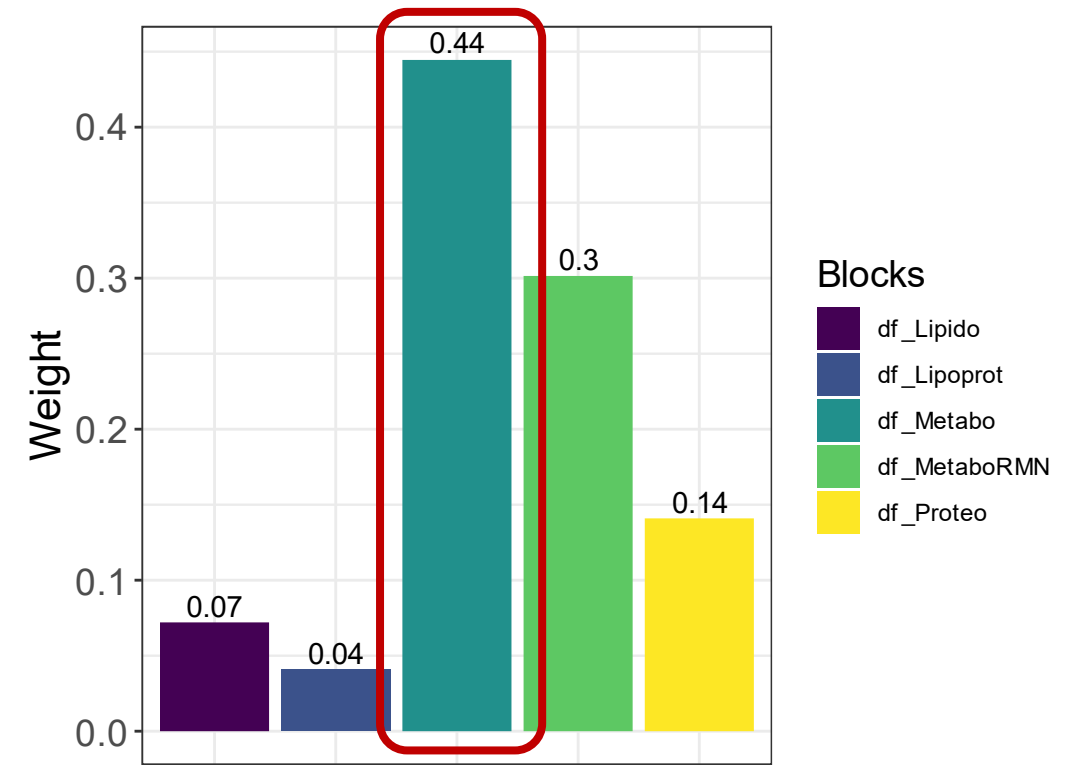
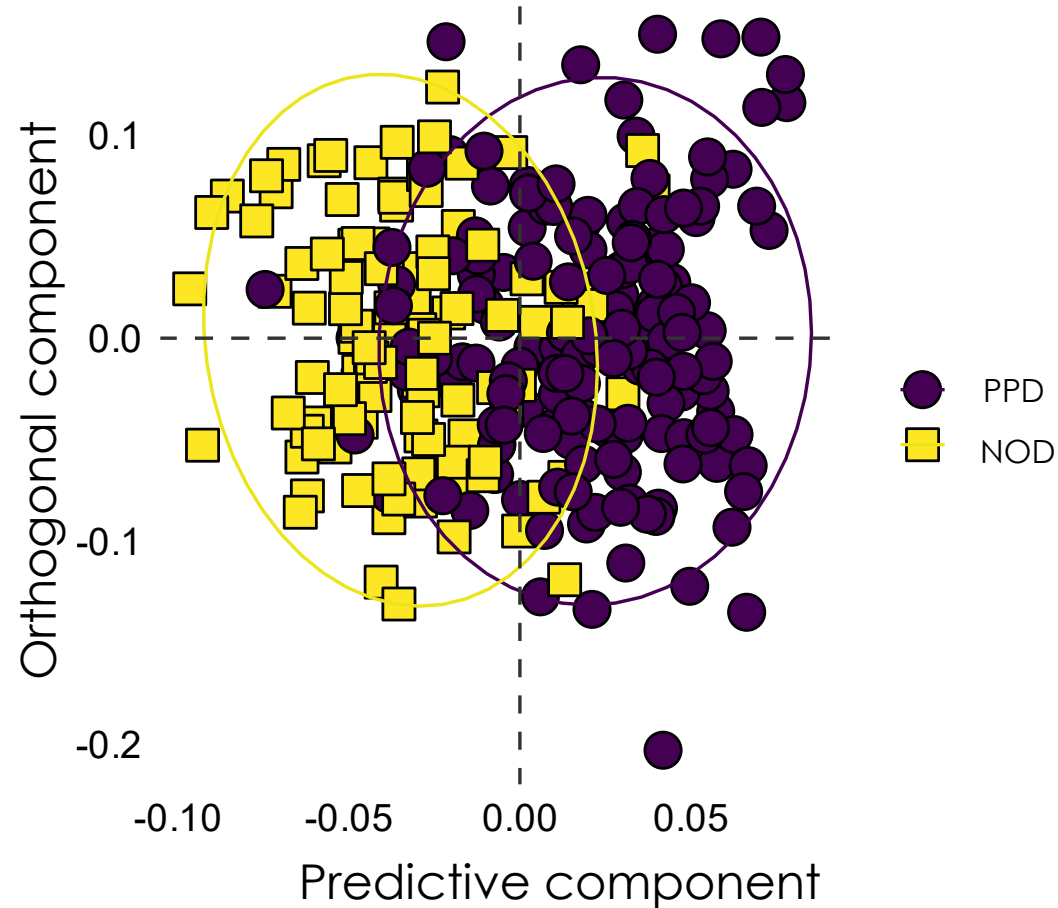
Samples collected at baseline (Vi) analysis



→ The metabolites detected by **NMR** are the main contributors to the discrimination between the groups

Analyse multi-omique et pré-sélection des biomarqueurs

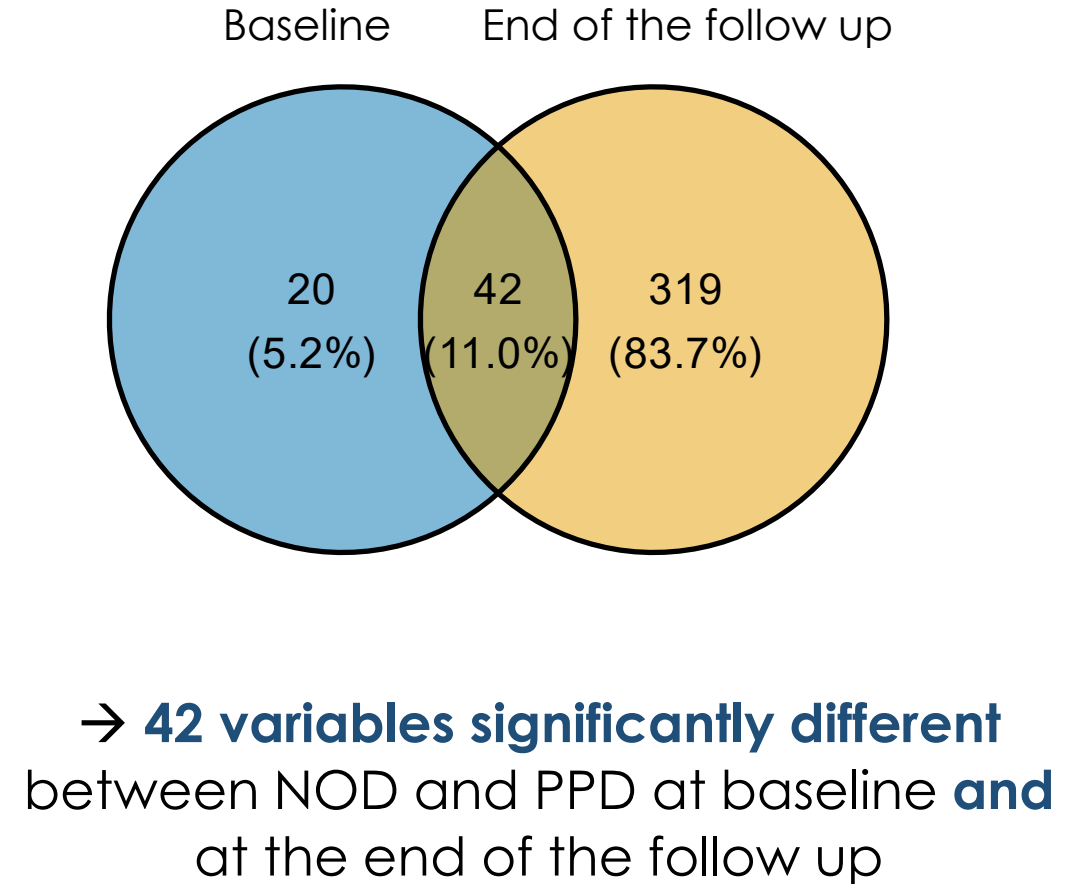
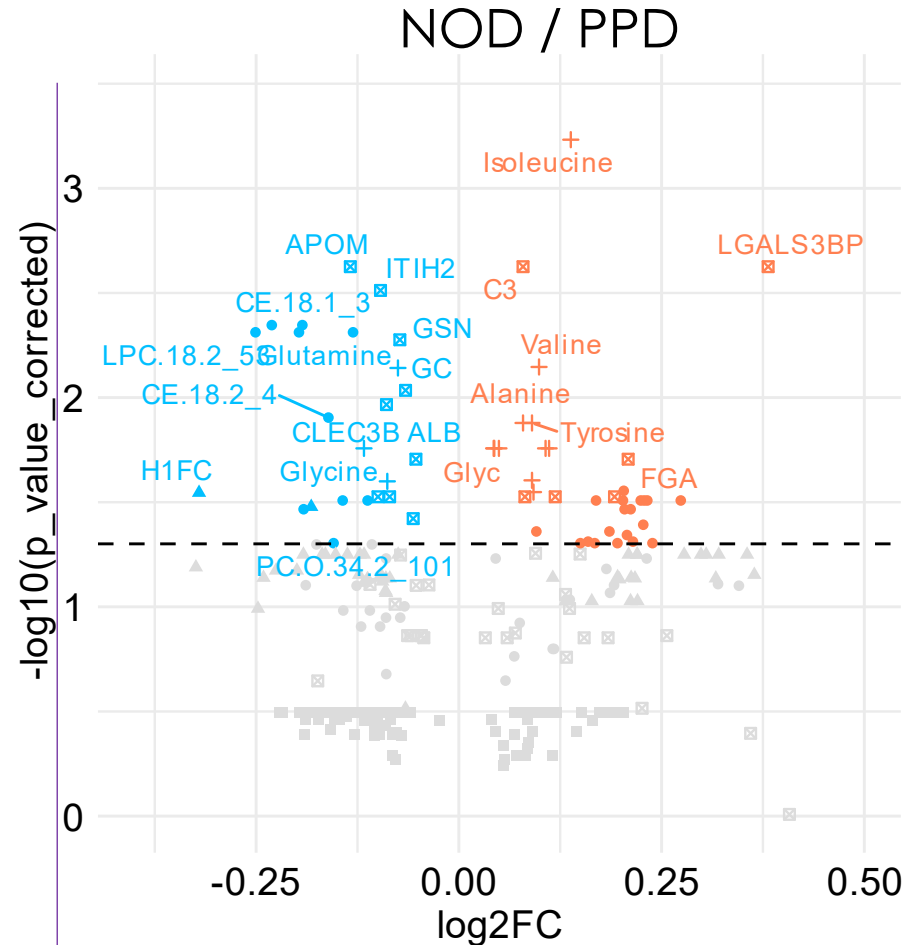
Samples collected at the end of the follow-up (Vf) analysis



→ Global metabolic changes following the transition to T2D

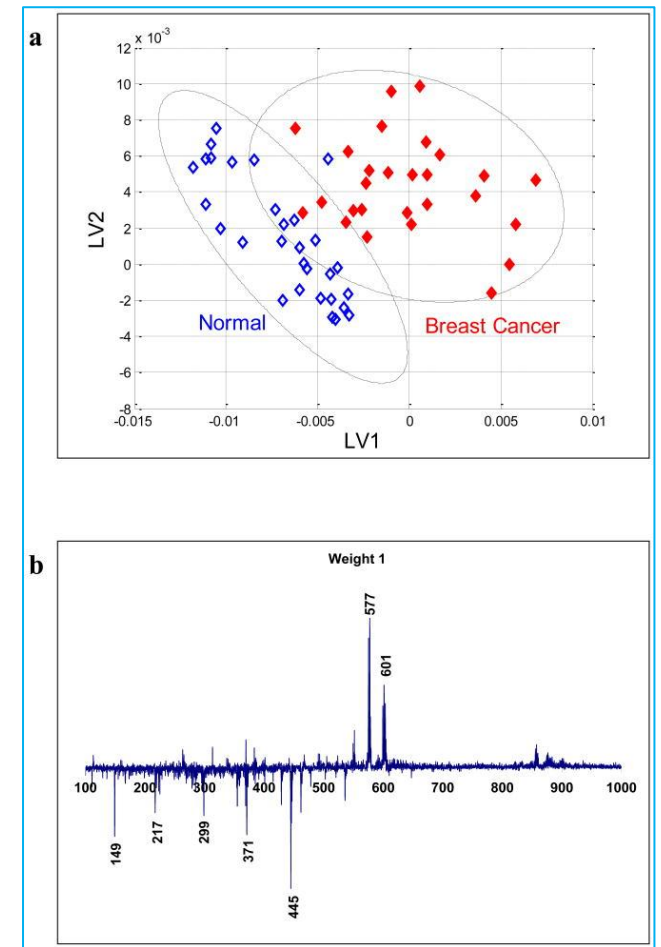
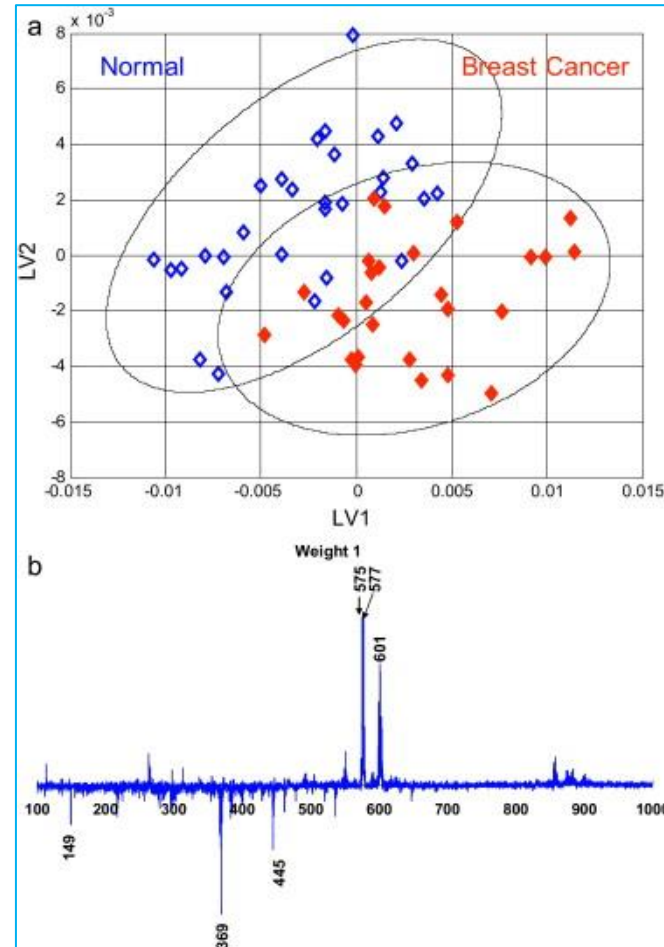
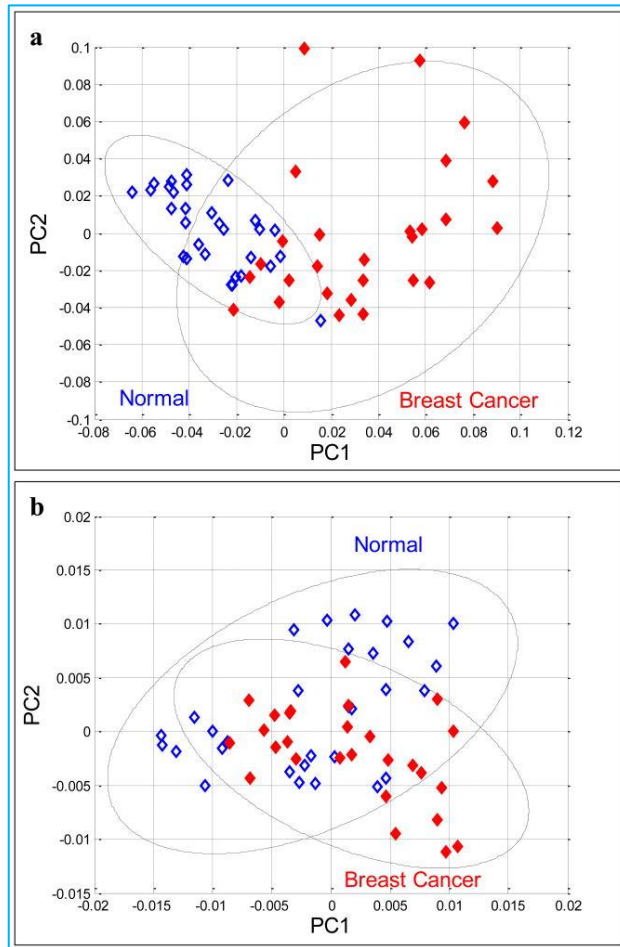
1008 variables
N-fold (n=10) cross validation
p value = 0.002 (Permutation test, n=500)

Analyse multi-omique et pré-sélection des biomarqueurs



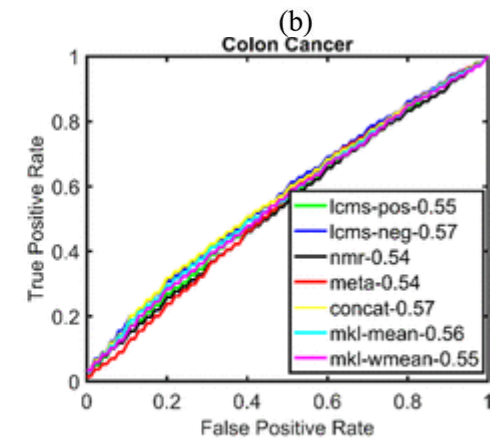
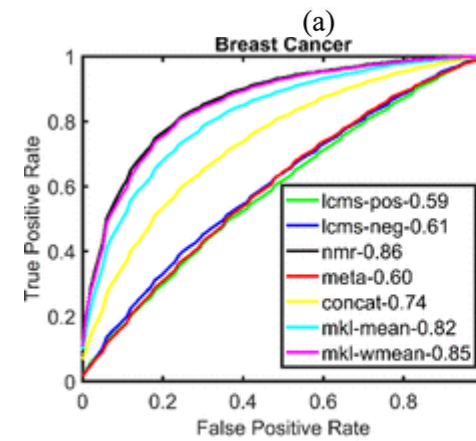
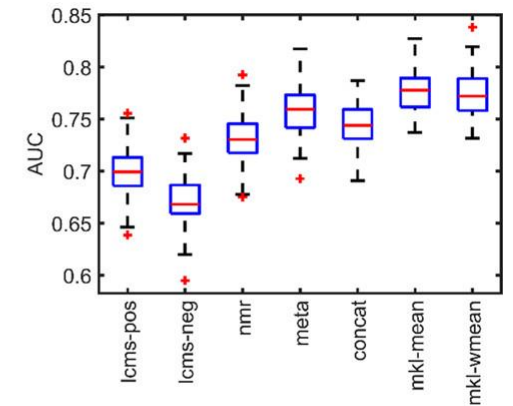
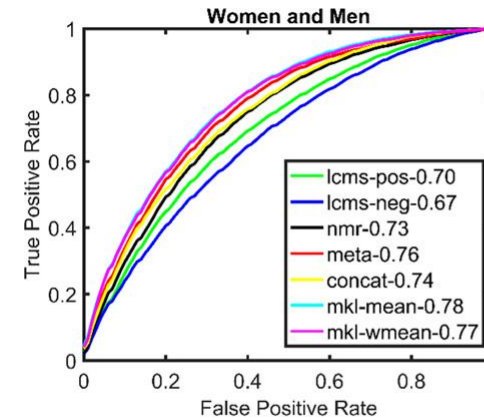
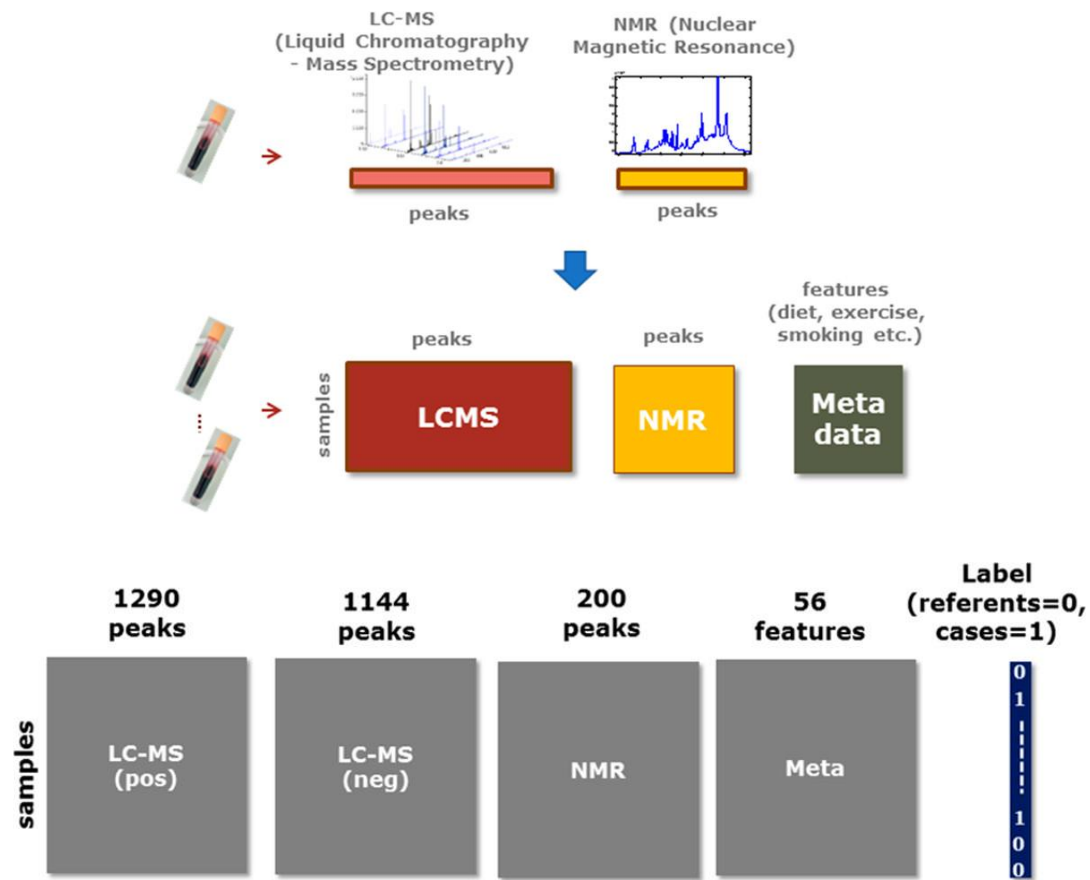
*Non-paired wilcoxon test or t test
FDR correction*

D'autres exemples de la littérature #1



D'autres exemples de la littérature #2

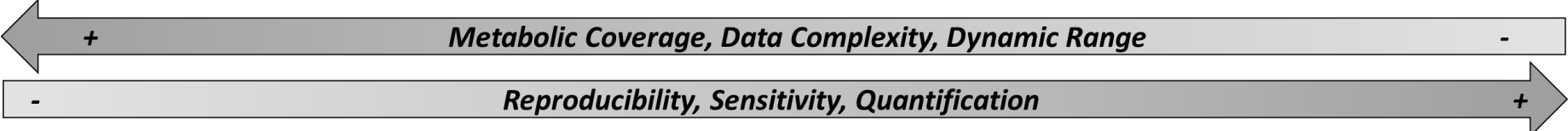
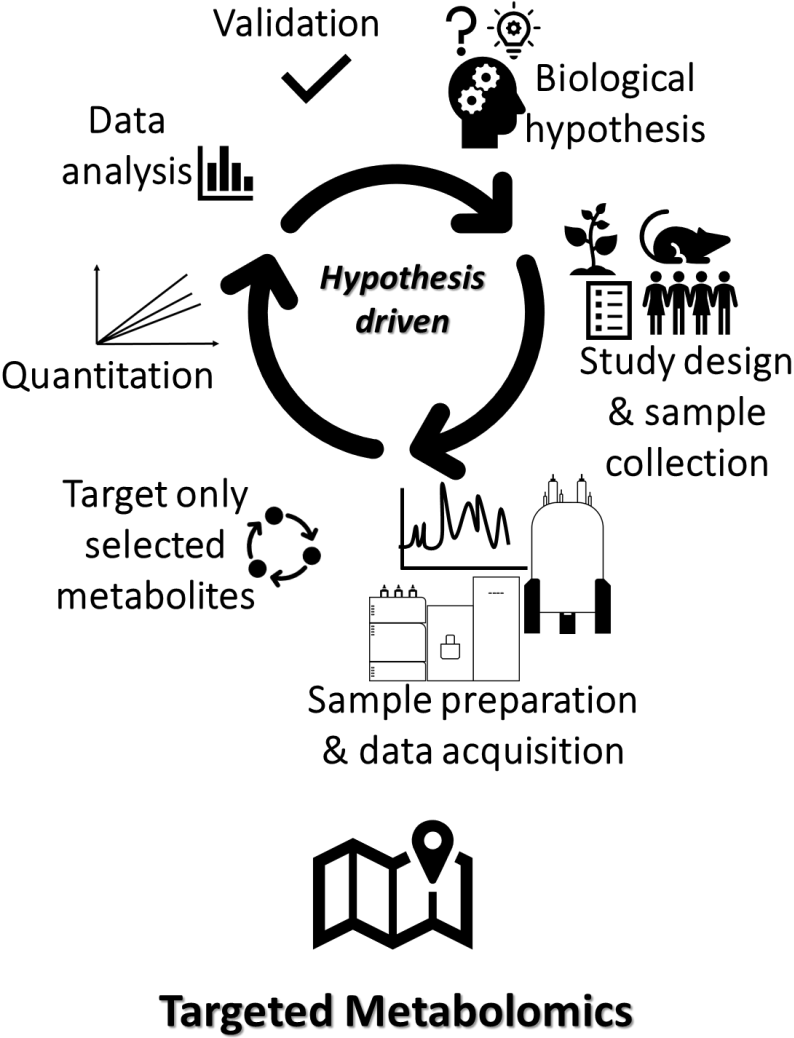
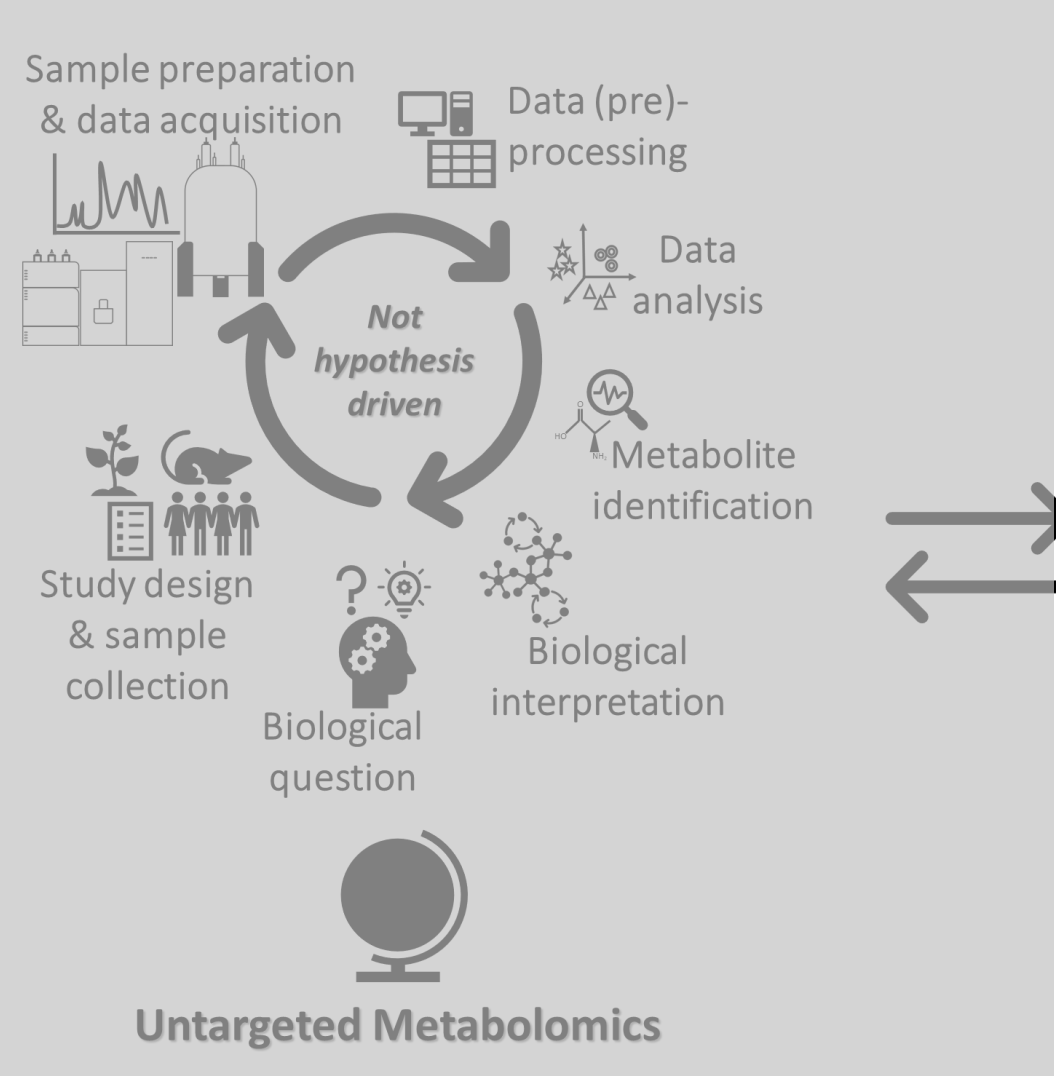
L'intégration MB n'est pas un outil infallible



(a)

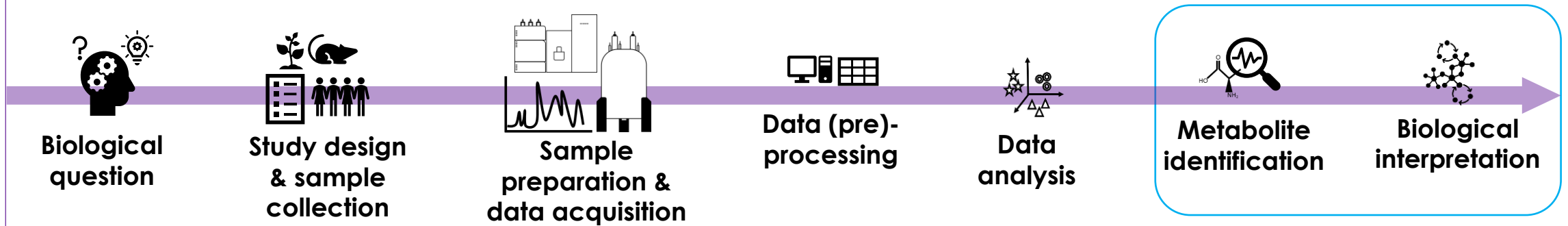
(b)

Dans le contexte d'une approche ciblée



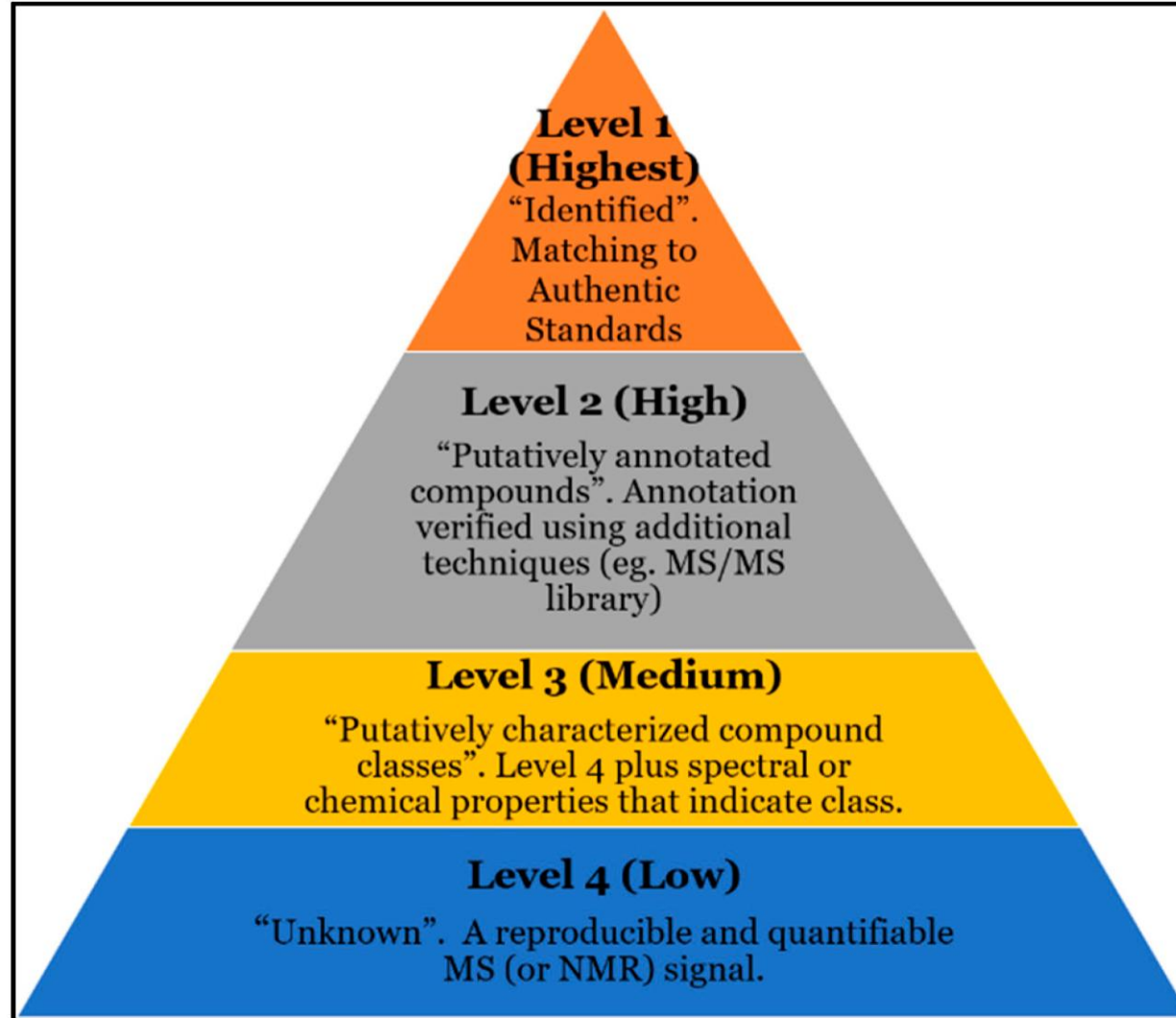
Combined NMR spectroscopy and MS applied to metabolomics

- Combination following a metabolomics analysis
 - To help the identification of metabolites / potential biomarkers
 - To quantify
 - Fluxomics



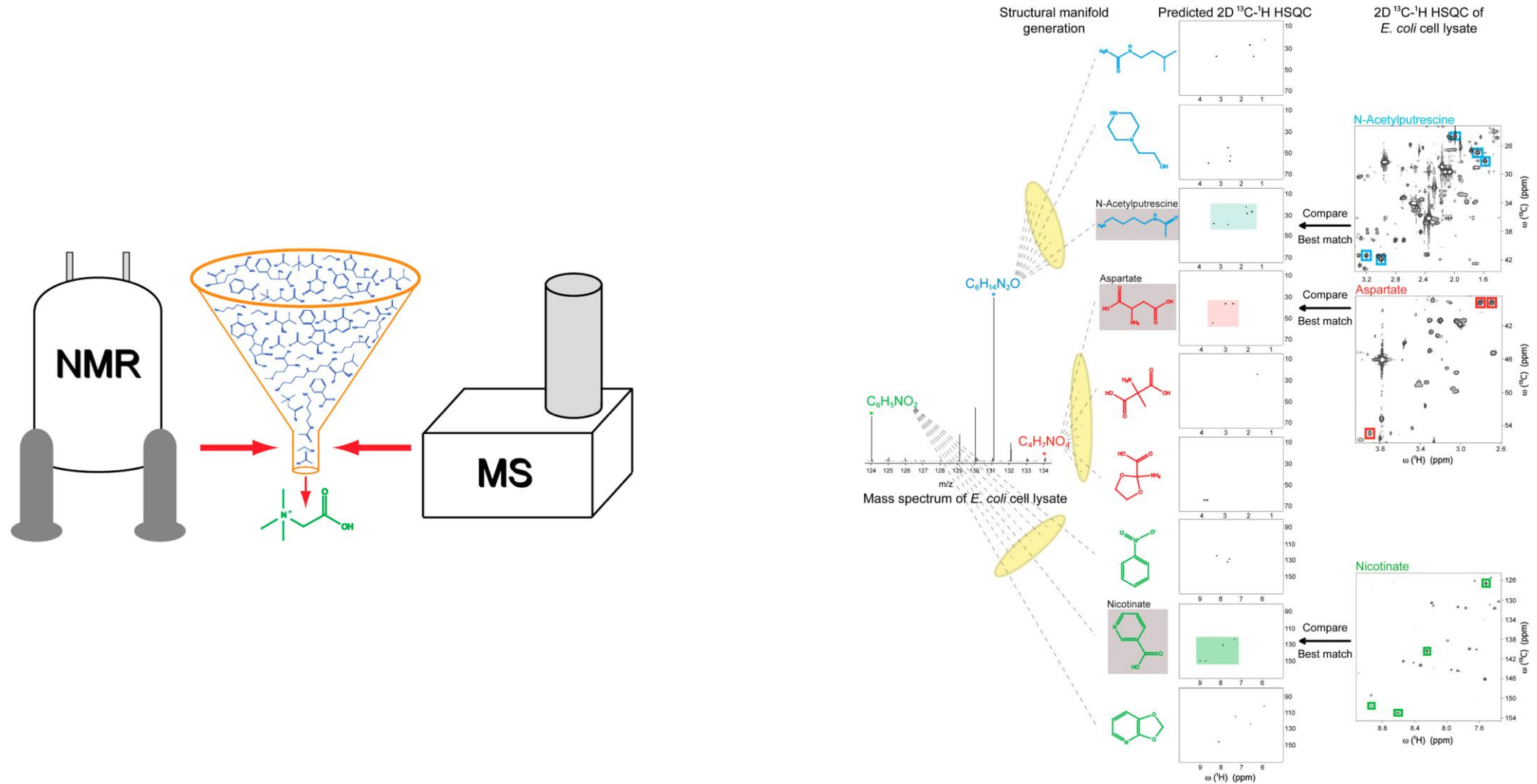
Combination following a metabolomics analysis

To help the identification of metabolites / potential biomarkers



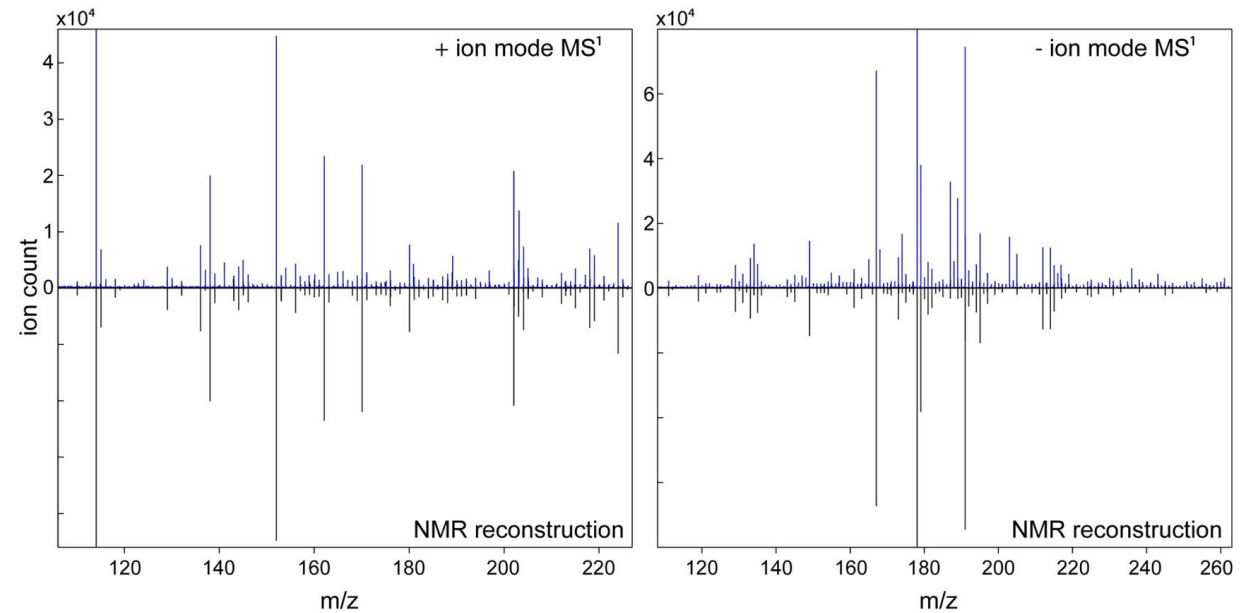
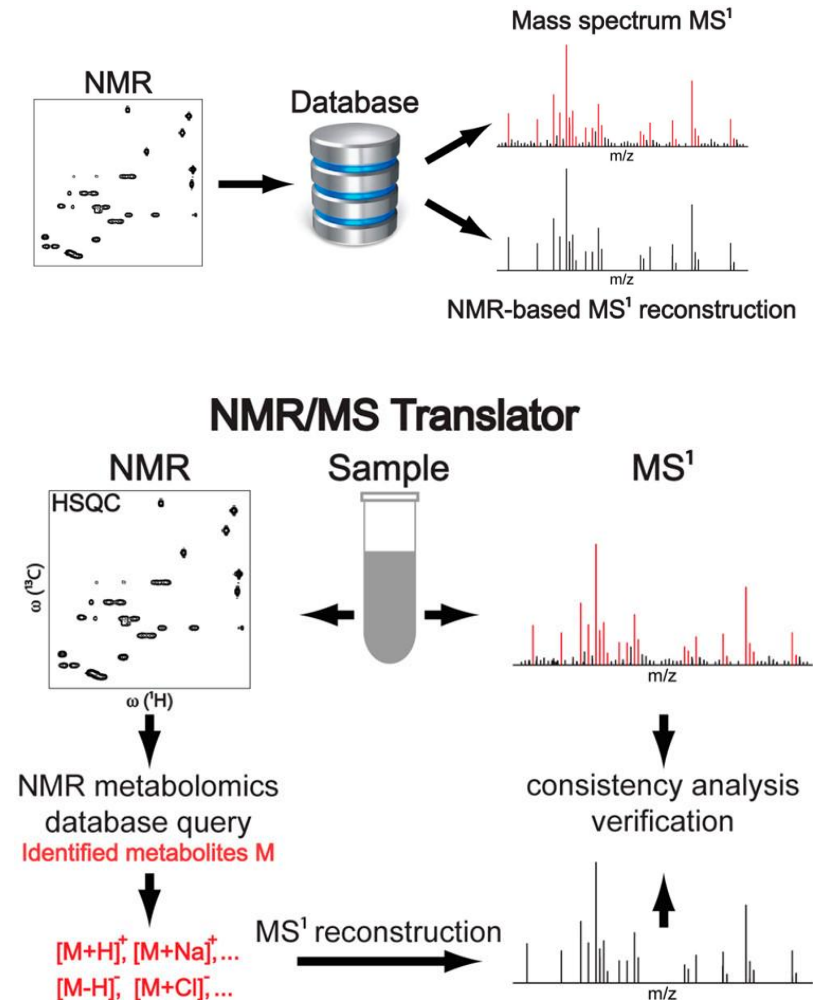
Combination following a metabolomics analysis

To help the identification of metabolites / potential biomarkers



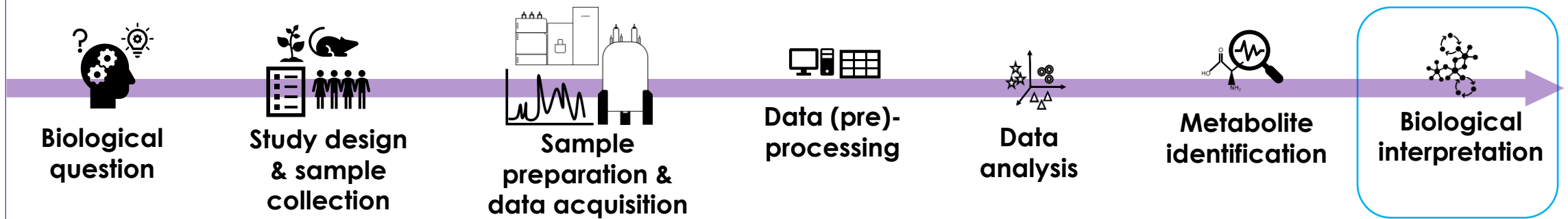
Combination following a metabolomics analysis

To help the identification of metabolites / potential biomarkers



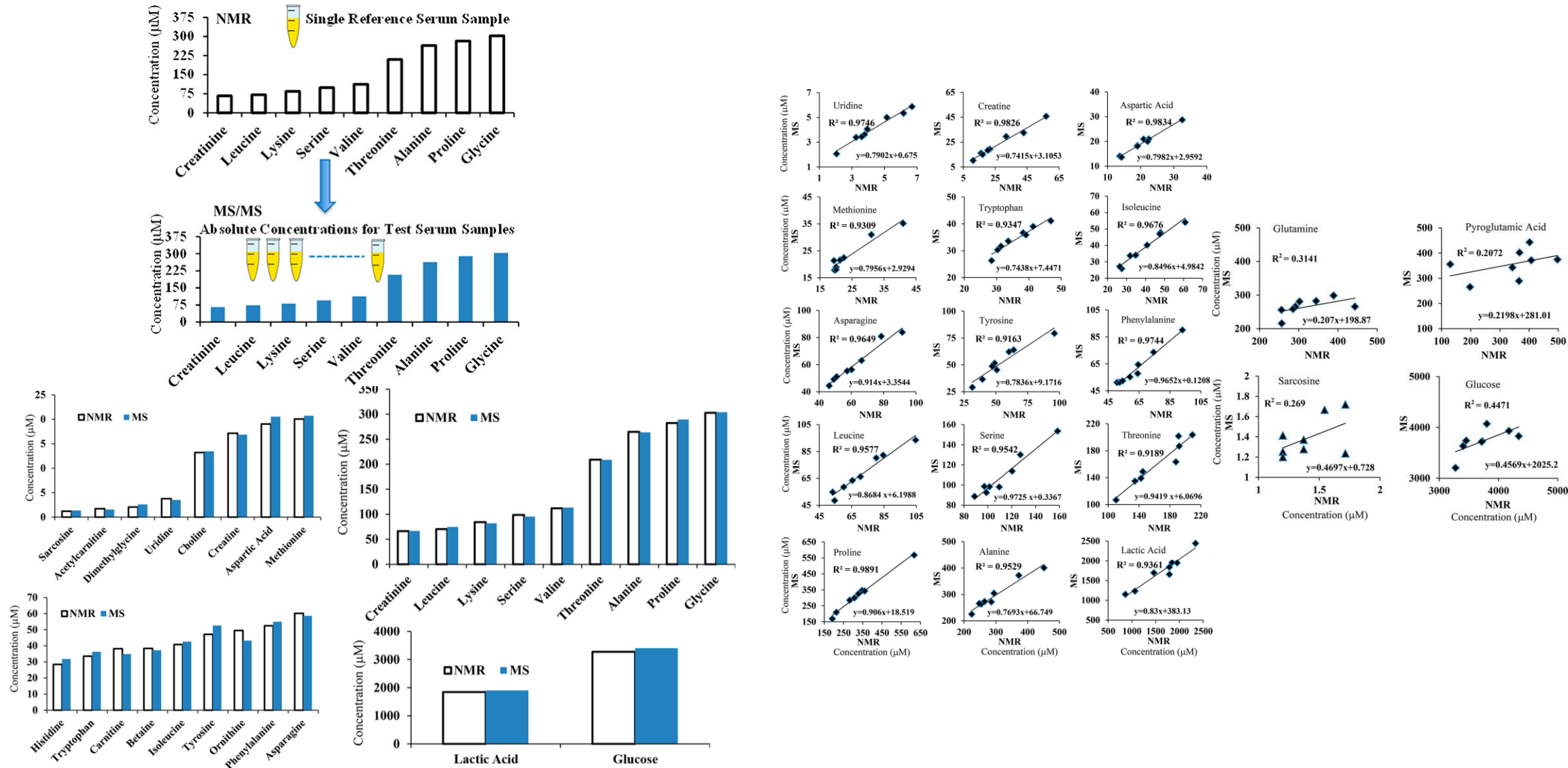
Combined NMR spectroscopy and MS applied to metabolomics

- Combination following a metabolomics analysis
 - To help the identification of metabolites / potential biomarkers
 - To quantify
 - Stable-Isotope Resolved Metabolomics (SIRM) or Fluxomics



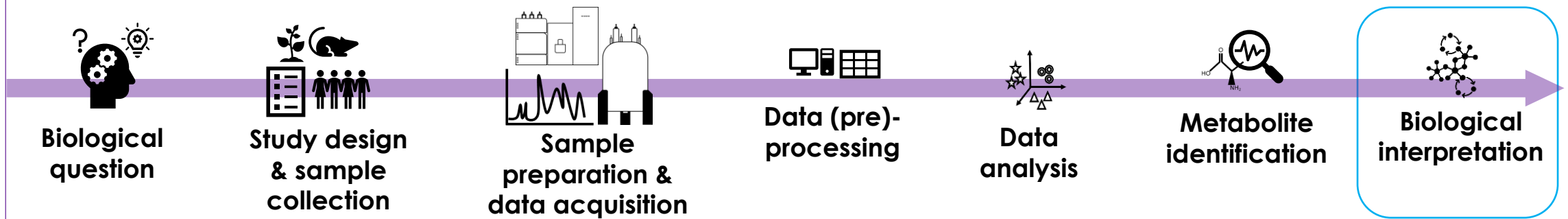
Combination following a metabolomics analysis

To quantify



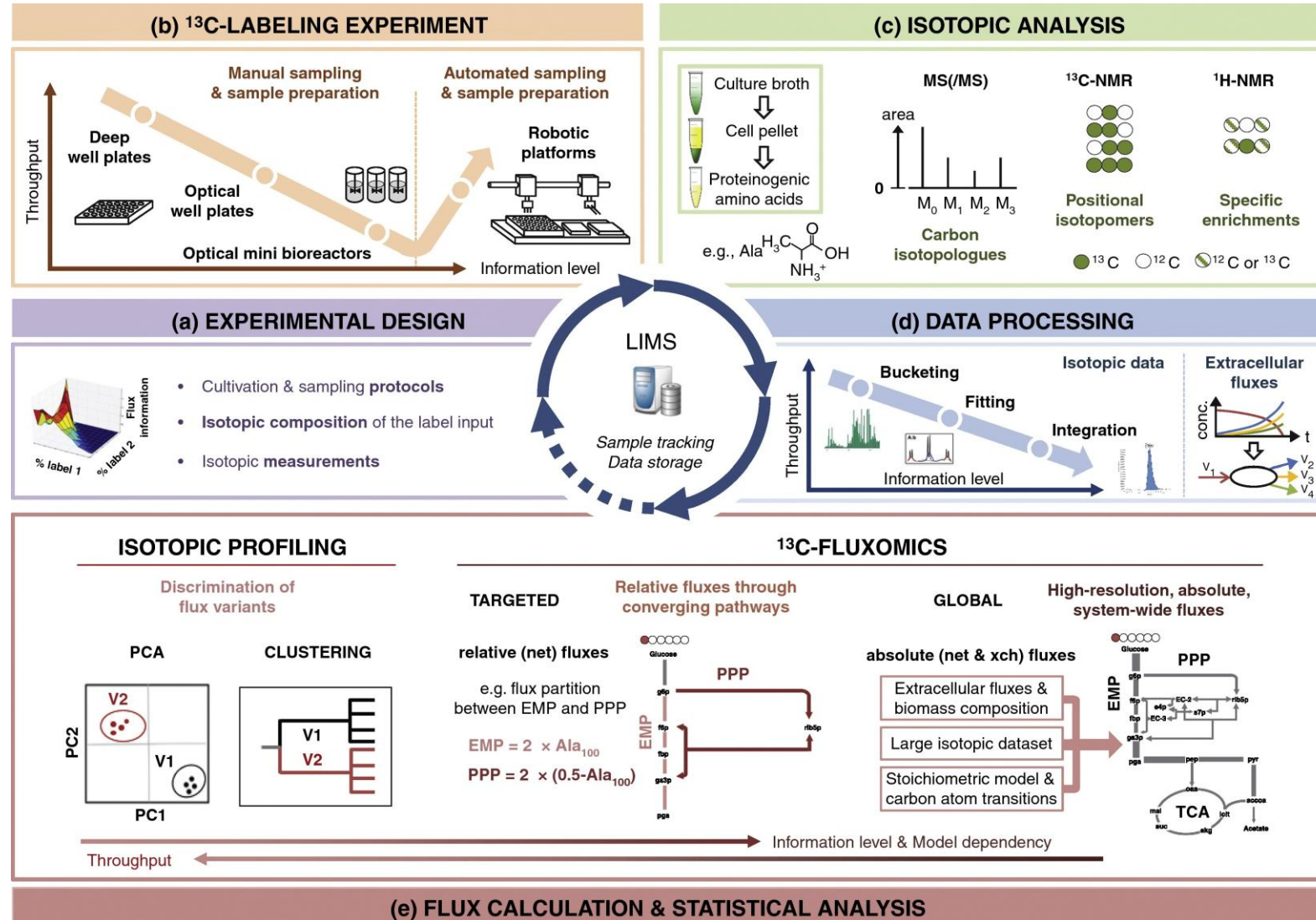
Combined NMR spectroscopy and MS applied to metabolomics

- Combination following a metabolomics analysis
 - To help the identification of metabolites / potential biomarkers
 - To quantify
 - Fluxomics



Combination following a metabolomics analysis

Fluxomics



Conclusion

- Many different applications in metabolomics
- Each step of the workflow must be performed carefully and in the same way for all samples
- To overcome current bottleneck of metabolomics, the combine use of NMR and MS can help, thanks to:
- Data sets combination
 - Cross-comparaison to increase metabolic coverage → Automatically done when correlation or MB is aimed
 - Correlation of NMR and MS data sets → Computational improvements is helping
 - Multiblock fusion / integration → Computational improvements is helping
- Combination following a metabolomics analysis
 - To help the identification → Niche to obtain a precise but crucial information
 - To quantify → Niche to obtain a precise but crucial information
 - Fluxomics → Computational improvements is and will keep helping

Acknowledgments

Equipe MIMM

Patrick Giraudeau
Jean-Nicolas Dumez
Virginie Silvestre
Estelle Martineau
Jonathan Farjon
Marine Piou
Victor Ribay
Pia Mayer
Benoît Charrier
Julien Gauvreau
Sumit Mishra
Martin Gualbert
Marie Delage
Jeremy Marchand
Arnab Day
Stéphane Guérin

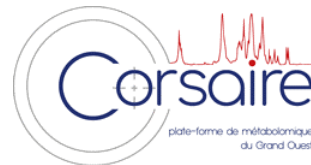


**UNIVERSITÉ
DE GENÈVE**

Julien Boccard



Claire Cannet



Chloé Cloteau
Amandine Rocher
Mikaël Croyal
Bertrand Cariou
Samy Hadjadj
Claire Wegner
Matthieu Wargny
Manon Guibot
Emilie Charon
Mathilde Gourdel
Arsênio Rodrigues

